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PILM'25

Proceedings of the International Workshop on
Personalized Incremental Learning in Medicine

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Simone Palazzo (University of Catania, Italy)
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**Association for
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PILM 2025 Chairs' Welcome

It is our pleasure to welcome you to the 2025 ACM Workshop on Personalized Incremental Learning in Medicine (PILM), held in conjunction with ACM Multimedia 2025. This workshop aims to explore the critical intersection of incremental learning and multimedia applications within the healthcare domain. The mission of PILM is to create a forum for discussing how recent advancements in continual and incremental learning can be applied to personalized medicine, leveraging data such as medical imaging, patient records, and biometric sensor data. We seek to bridge the gap between theoretical frameworks and their practical applications, focusing on adapting models to new information while retaining previously learned knowledge—a crucial capability for tailoring treatments to individual patient characteristics and overcoming challenges posed by high-dimensional, sparse, and heterogeneous data sets.

The call for papers attracted high-quality submissions. Each paper was carefully reviewed by two members of our program committee to ensure a strong and engaging technical program. Overall, the workshop received 6 regular paper submissions, 4 of which were accepted (66.7%).

The success of PILM 2025 is the result of the collective effort of many individuals. Our primary thanks go to the authors, whose innovative research provides the foundation for our technical program. We are also deeply indebted to our program committee members for their insightful reviews. A special thank you is also due to the ACM Multimedia 2025 organizing committee for their support and for hosting our workshop.

The workshop chairs,

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PILM 2025 Workshop Organization

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A Peer-to-Peer Federated Incremental Learning Framework for Multi-Institutional Low-Dose CT Denoising

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Abstract

Deep learning techniques have been widely used in computed tomography (CT), but they require large datasets to train networks. Moreover, data sharing among multiple institutions is limited due to data privacy constraints, which hinders the development of high-performance DL-based CT imaging models that rely on multi-institutional collaborations. Federated learning (FL) is an alternative strategy for training models without centralized data across multiple institutions. In this work, we propose a novel peer-to-peer federated incremental learning strategy to improve low-dose CT imaging performance across multiple institutions. The newly proposed method is called peer-to-peer incremental FL with intermediate controllers, i.e., icP2P-FL. Specifically, unlike conventional FL models, the proposed icP2P-FL does not require a central server to coordinate training information for a global model. In the icP2P-FL method, peer-to-peer federated incremental learning is introduced, where each institution jointly maintains an institution-shared sub-model with prompt learning, which shares network parameters exclusively with each decoder. The framework incorporates decoded features from the CT images of all institutions via peer-to-peer cyclic information exchange. Furthermore, an intermediate controller with self-aware regulation is designed to enhance

the flexibility of overall training. Extensive experiments were conducted on the AAPM Low-Dose CT Grand Challenge dataset and local datasets. The results demonstrate that the proposed icP2P-FL method outperforms other comparative approaches both qualitatively and quantitatively, achieving accuracy comparable to that of a model trained on centralized data from all institutions.

CCS Concepts

• **Computing methodologies** → **Cooperation and coordination**; **Reconstruction**.

Keywords

Low-Dose CT, Peer-to-peer Federated Learning, Incremental learning, image denoising

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1 Introduction

Low-dose computed tomography (CT) imaging has been widely used in clinics. However, lowering the dose level can result in degraded image quality [12]. Deep learning (DL)-based methods have been developed to improve low-dose CT image quality, including data-driven DL methods [19, 25] and model-driven deep unrolling methods [6, 7]. However, most DL-based methods for low-dose CT imaging usually require large amounts of diversity-rich data which can be labor-intensive to collect. In addition, it is difficult to collect and share CT data from multiple institutions efficiently due to patient privacy [14]. A single-institution training network may

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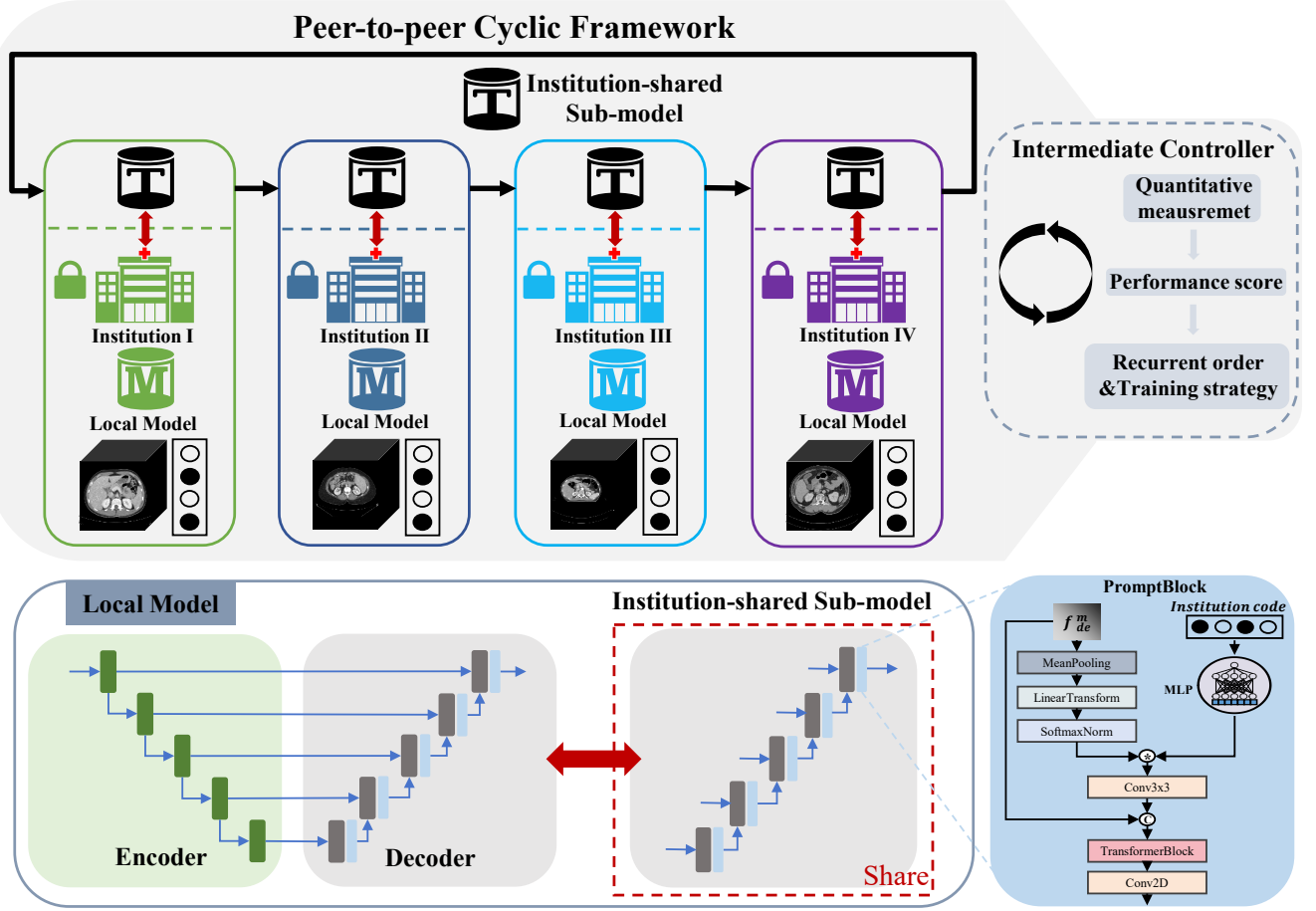


Figure 1: The framework of the presented icP2P-FL.

suffer from inadvertent bias that ultimately limits imaging performance even on the data collected from same institution due to the considerable heterogeneity exists among multiple institutions.

Federated learning (FL) enables decentralized model training across geographically dispersed multi-institutional data silos to address these limitations. In the FL, distributed institutions collaboratively learn a shared model while keeping data local for privacy [16]. The FL is also used in the medical imaging field and obtains better benefits over traditional DL-based methods [4, 5, 21, 24]. In centralized (center-to-peer) federated learning, a central server is required to orchestrate the process of training a single shared model, which needs high communication costs between the central server and multiple institutions. Various strategies have been investigated to reduce communication costs [10, 18]. In particular, decentralized FL (peer-to-peer) obviates the need for a central server and more effectively addresses challenges inherent to centralized FL models [17, 20]. In the peer-to-peer (P2P) FL model, to avoid the catastrophic forgetting problem, incremental learning plays a critical role [2, 8, 11]. The incremental learning strategy in the P2P FL can train the model with the ability to remember old knowledge and learn new knowledge.

In this work, inspired by the P2P FL and incremental learning strategy, we propose a novel P2P federated incremental learning strategy to improve low-dose CT imaging performance from multiple institutions. The newly proposed method is termed as P2P incremental FL with intermediate controllers, i.e., icP2P-FL. To the best of our knowledge, this is the first one to utilize P2P incremental FL in low-dose CT image reconstruction. Specifically, the proposed icP2P-FL does not require a central server and adopts a P2P model where the institutions only communicate with their one-hop neighbors to reduce certain communication costs. In the proposed icP2P-FL method, due to the stable and effective common features among different institutions, all the institutions collaboratively share an institution-shared sub-model with prompt learning and directly interact with each other without depending on a central server. The framework incorporates decoded features from the CT images of all institutions via peer-to-peer cyclic information exchange. During training, each institution first uses the institution-shared model with prompt block to extract decoded features from its local low-dose CT images—these features contain key information such as anatomical structure details, noise distribution patterns, and tissue attenuation characteristics. In this process, it shares its

locally generated decoded features while receiving features from neighboring institutions to update its own decoder parameters. This cyclic exchange of features ensures that information from all institutions continuously propagates through the network, enabling the model to capture cross-institutional commonalities and adapt to diverse data distributions across different centers. Each institution can initiate an update process dynamically. Furthermore, an intermediate controller is developed to make the overall training more flexible. Due to the high frequency of interaction and the intermediate controller, the proposed icP2P-FL converges quickly. The CT dataset with different protocols from multiple institutions is collected to validate and evaluate the denoising performance of the proposed icP2P-FL for image reconstruction. Experimental results show that the proposed icP2P-FL method obtains competitive denoising performance in the CT reconstruction task compared with the other competing methods, and reaches an accuracy similar to a model trained with centralized data from all the institutions.

2 Methods

Let us consider an environment with K institutions, where each institution has training data $D^k = \{x_i^k, y_i^k\}_{i=1}^{n^k}$ and characteristic data $C^k = \{z_i^k\}_{i=1}^{n^k}$ with n^k labeled samples. Illustrated by Figure 1, icP2P-FL consists of the institution-specific local model, peer-to-peer cyclic framework, and intermediate controller. In the peer-to-peer cyclic framework of icP2P-FL, there is no central server to coordinate training; instead, institutions are peer-to-peer transmitted via institution-shared sub-model parameters with prompts, enabling incremental learning of decoded features and preserving personalized knowledge. The intermediate controller adaptively adjusts framework strategies to prevent catastrophic forgetting.

2.1 The Overview of icP2P-FL

For simplicity, we focus on four institutions in this paper but icP2P-FL can be easily extended to more institutions. Figure 1 shows the framework of four institutions. The presented icP2P-FL method consists of a peer-to-peer FL framework, an intermediate controller, and four local models for each institution. The local model employs a UNet architecture for its exceptional performance in image denoising tasks. UNet's unique design, featuring a contracting path to capture context and a symmetric expanding path for precise localization, makes it highly effective in reconstructing detailed images from input data. Its ability to handle varying image scales and maintain spatial information integrity is particularly beneficial in medical imaging and other segmentation-intensive applications. Additionally, UNet's flexibility allows for adaptation to different network depths and types, making it suitable for the diverse and potentially heterogeneous data environments encountered in icP2P-FL's peer-to-peer framework. The training process of one cycle is conducted as the following steps:

- (1) The initial institution order is set (Initial_sequence: [Institution I, Institution II, Institution III, ..., Institution k]), and network parameters are initialized.
- (2) The dataset D^1 of Institution I is used to train the local model $\Omega_1(\cdot; \omega_1)$ for CT image denoising.

- (3) After the training procedure is completed in Institution I, the trained model $\Omega_1(\cdot; \omega_1)$ is employed to evaluate the performance on the characteristic data from Institution I to obtain quantitative measurements, i.e., $[p, s, m]_I^k$.
- (4) The parameters ω_1^s in the institution-shared sub-model Ω_1^s are transferred to the one-hop neighbor, i.e., Institution II. D^2 is used to fine-tune the institution-shared sub-model with prompt learning to retain the global feature and archive the catastrophic forgetting compensation.
- (5) After the training is completed in Institution II, the trained model $\Omega_2(\cdot; \omega_2)$ is also employed to evaluate the performance on the characteristic data from Institution II to obtain quantitative measurements, and the other institutions are trained similarly.
- (6) When all the institutions finish training, the intermediate controller grades all the quantitative measurements $[p, s, m]$, of all the institutions on-line. This will immediately determine the inter-institutional training strategy and whether the global shared model is well-trained.

It should be noted that the effect of the intermediate controller with adaptive adjustment can help icP2P-FL train the personalized local model more flexibly and efficiently. After completing multiple cycles, icP2P-FL can obtain the well-trained local model to obtain high-quality CT images from all the institution efficiently.

2.2 Inter-institutional Incremental Learning

In this study, we extend the cyclic institutional incremental learning (CIIL) framework to preserve robust model performance across multiple institutions in the transmission cycles. In institutional incremental learning (IIL), multiple institutions continuously train a shared model in a set order, with each institution training the model once per cycle before aggregation into the final model. Moreover, cyclic institutional incremental learning (CIIL) extends this process by repeating these IIL cycles across institutions and reducing forgetting through the careful determination of the number of training rounds at each institution.

Referring to the CIIL framework [3, 22], the proposed icP2P-FL method adopts the personalized strategy to keep global knowledge of previous institutions and learn personal knowledge for new institution. Specifically, we divide the UNet of each institution into two sub-models, i.e., the personalized sub-model (encoder) Ω_k^p and the institution-shared sub-model (decoder) Ω_k^s , and k -th institution's local model Ω_k can be expressed as follows:

$$\Omega_k(x_i^k; \omega_k) = \Omega_k^s \left[\Omega_k^p(x_i^k; \omega_k^p), \text{skip}_k, \lambda_k; \omega_{k-1}^s \right], \quad (1)$$

where x_i^k is the i th low-dose CT image, y_i^k is the normal-dose CT images data in each institution dataset D^k in a batch, respectively. ω_{k-1}^s is the transmitted parameter of the previous institution after completing all training of the institution-round. skip_k represents the set of skip-connections bridging the personalized encoder Ω_k^p and the shared decoder Ω_k^s . λ_k is the institution code for the k -th institution, used to prompt the outputs of decoder according to institutional characteristics.

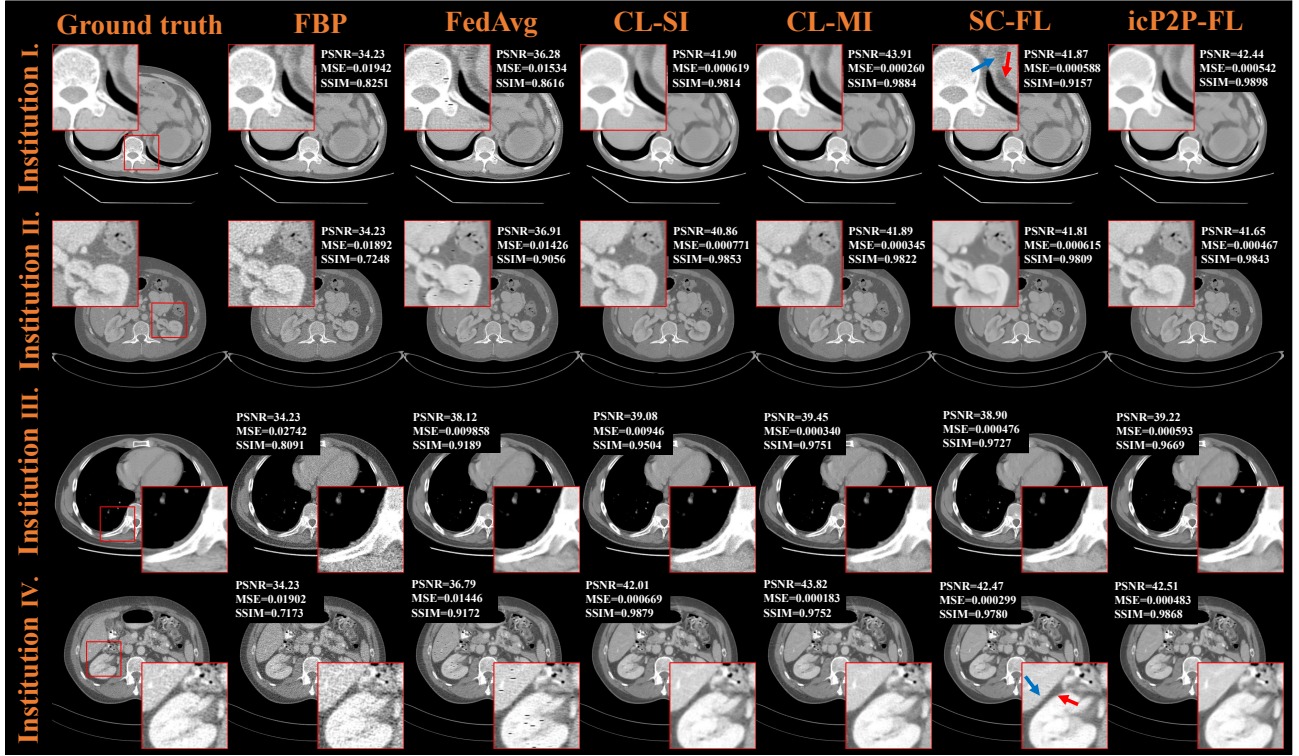


Figure 2: Qualitative comparison of the image reconstructed by the different methods. The display windows for CT images at Institution 1, Institution 2, Institution 3 and Institution 4 are $[-50, 50]$, $[-100, 100]$, $[-50, 50]$, $[-150, 150]$ HU, respectively. The display windows for zoomed-in ROIs are $[-10, 50]$, $[-20, 80]$, $[-30, 50]$, $[-80, 100]$ HU, respectively.

In the t -th round, the local model parameters at the k th institution can be updated as follows:

$$\omega_k^{(t+1)} = \omega_k^{(t)} - \eta \cdot \nabla_{\omega_k^{(t)}} \mathcal{L}_2, \quad (2)$$

where η denotes the learning rate at each institution. $\nabla_{\omega_k^{(t)}} \mathcal{L}_2$ represents the gradient of the loss function with respect to $\omega_k^{(t)}$.

$\mathcal{L}_2 = \left\| \Omega_k(x_i^k; \omega_k^t) - y_i^t \right\|_2^2$ is the mean squared error (MSE) loss.

2.3 Intra-institutional Prompt Learning

To further enhance the institution-shared sub-model’s ability to generalize across different institution, we introduce a prompt learning block (PromptBlock). The PromptBlock dynamically generates prompt components based on the target institution code λ_k , which are then used to guide the feature extraction process in the decoder. At each convolutional block in the decoder, the prompt learning module consists of two main steps: prompt generation and feature fusion. First, the institution kernel code λ_k is parameterized into prompt components using a Multi-Layer Perceptron (MLP), which are then combined with the feature map to generate task-specific prompts that enhance feature representation. The prompt generation process can be expressed as:

$$P = \text{Linear}(f_{de}^m) \odot \text{MLP}(\lambda_k), \quad (3)$$

where $\odot$ represents element-wise multiplication, Linear denotes a linear conversion that aligns the dimensions of the prompt components $\text{MLP}(\lambda_k)$ with multi-scale decoder features f_{de}^m . $P \in \mathbb{R}^{N \times M}$ is the generated prompt, N is the number of prompt components, and M is the channel dimension of the feature map. Next, P is processed through a convolutional layer, and then fused with f_d . The fusion process can be described as:

$$\tilde{f}_{de}^m = \text{Transformer}(\text{Concat}(f_d, P)), \quad (4)$$

where $\tilde{f}_{de}^m$ is the enhanced feature map obtained by processing the concatenation of f_{de}^m and P through a Transformer block [22]. This operation integrates institution-specific guidance from P into the feature representation, enabling more effective feature adaptation.

2.4 Intermediate Controller

As the number and distribution of training datasets in each institution varies a lot, there are real differences in the computational efficiency and catastrophic forgetting rate of training the shared global model on different datasets. The scores can be obtained from the quantitative measurements $[p, s, m]$, which p , s , m denotes respectively peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and mean square error (MSE). These metrics evaluate the denoising performance on each institution’s characteristic dataset. The intermediate controller sorts all the scores of

the institutions online in descending order, and then immediately determines inter-institutional training order (e.g., [Institution I, Inst. II, Inst. III, Inst. IV] is adjusted to [Institution IV, Inst. I, Inst. III, Inst. II]). Moreover, by the scores of the entire dataset of each institution, the intermediate controller can adjust the number of transmissions between two adjacent institutions (i.e., the number of cycles in the training stage) to balance learning and forgetting progress. And it also determines the institution-round at each institution (i.e., the number of training rounds in one cycle) according to each institution's performance trends. Finally, it can check whether each local model is well-trained against a predefined performance threshold and schedule additional cycles or rounds if necessary. This adaptive process enables rapid convergence and minimizes catastrophic forgetting across heterogeneous datasets. It is worth noting that the flexibility of the CRL framework extends to allowing non-participating external institutions to participate at any stage, enabling smooth integration of additional contributors. Similarly, participating institutions retain the autonomy to withdraw from the collaborative effort post-network updates, ensuring adaptability to evolving needs and preferences.

2.5 Implementation Details

In this work, the backbone DL network of the presented icP2P-FL method adopts the same network structure UNet [15], which contains an encoder and a decoder. The encoder includes four 4×4 convolution operations with stride 2 for downsampling, while the decoder performs the opposite operation, using four 2×2 ConvTranspose2d layers with stride 2 for upsampling. The TransformerBlock in PromptBlock follows the design and hyperparameters outlined in [22]. The training parameters are set as follows: (1) the number of transmissions and the number of institution-round of each institution are respectively set to 10 and 5, (2) the training round number is set to 100 and the weight is decayed at the 100th round by multiplying 0.2, (3) the learning rate and batch size of the each institution are 1.0×10^{-4} and 64, respectively, (4) all the institutions have paired data, i.e., normal-dose CT images/corresponding low-dose ones acquired with different protocols, (5) determination threshold is set to 1.3433. The FBP algorithm is based on the ASTRA toolbox. All networks were implemented using the PyTorch library, and their loss functions were optimized using the Adam optimizer algorithm. And all networks were trained on one NVIDIA RTX 4090 graphics processing unit (GPU) which has 24 GB memory capacity.

The presented icP2P-FL method is compared to several widely used denoising algorithms for image reconstruction, including the FBP algorithm, FedAvg [13], CL-SI (i.e., single institution model with centralized learning trained on institution-specific dataset), CL-MI (i.e., multiple institutions model with centralized learning trained on all datasets), and Semi-centralized Federated Learning network (SC-FL) [9]. Three measures, i.e., PSNR, SSIM, and MSE, are employed to quantitatively evaluate the performance of the proposed our proposed icP2P-FL method and the comparison models.

3 Experimental Results

3.1 Dataset

To validate and evaluate the denoising performance of the proposed icP2P-FL method for image reconstruction, four different CT

datasets are used in the experiments. Specifically, with the approval of the medical ethics committee of the local institution, three CT datasets were collected from three local hospitals, i.e., Institution I, Institution II, and Institution III. The three datasets are acquired with different protocols with different scanners. The fourth CT dataset is from the "2016 NIH-AAPM-Mayo Clinic Low Dose CT Grand Challenge" [1], termed as Institution IV. All the CT images are served as normal-dose ones in this study. Then we simulated low-dose CT images (i.e., quarter dose) from the corresponding CT images based on the previous study [23]. In Institutions I, II, and III, 800 and 100 cases are assigned for training and testing datasets, respectively. 200 and 100 cases are assigned for training and testing in Institution IV. To select characteristic datasets for validation evaluation, we employ 100 cases of paired images from each institution dataset independent of training and testing, with deeper features and individualized data distributions in each institution.

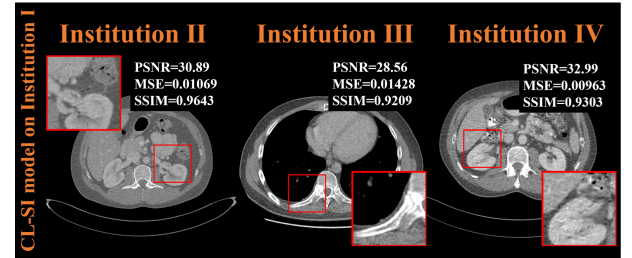


Figure 3: Results of institution II, III, IV tested by the CL-SI model on Institution I

3.2 Performance Comparison

Figure 2 shows the example slice denoising with each individual method at four institutions and a corresponding zoomed-in region-of-interest (ROI) view indicated by the red boxes. The zoomed-in ROI view provides enhanced visualization of the denoised image by revealing finer details. The normal-dose CT images are served as ground truth for comparison. It can be seen that the low-dose FBP images contain severe noise-induced artifacts. FedAvg can reduce noise-induced artifacts to some extent, but it introduces undesired artifacts in the final results. The main reason is that considerable heterogeneity exists among the four institutions, and FedAvg fails to fully take the considerable heterogeneity into consideration. The CL-SI is trained on the institution-specific CT data and it obtains promising results in noise-induced artifacts suppression, but it can not process low-dose CT images from other institutions shown in Figure 3. The degraded denoising performance for image reconstruction is attributed to the heterogeneity between the training dataset from one specific institution and the testing data from the other institutions. The CL-MI is trained on the centralized CT data from the four institutions, and it achieves the best performance in noise-induced artifacts removal and structure details recovery due to it considers all the latent characteristics in the four institutions. The SC-FL has better performance than the FedAvg and CL-SI by constructing high-quality labels in the central server. But it can be seen that the SC-FL results still contain residual noise-induced

Table 1: Mean and standard for PSNR and SSIM of institutions.

PSNR	Institution I	Institution II	Institution III	Institution IV
FBP	34.12±1.2814	34.43±1.3234	33.98±1.3848	34.93±1.3176
FedAvg	35.11±1.2679	36.39±1.0313	36.95±1.4068	36.79±1.0041
Single_institution	41.01±0.2628	39.96±0.2920	39.02±0.2182	41.75±0.3012
Multi_institutions	43.17±0.2619	41.83±0.3237	39.33±0.4146	42.91±0.2751
SC-FL	41.75±0.2679	42.09±0.3313	38.09±0.4068	42.64±0.3041
icP2P-FL	41.78±0.2628	42.55±0.2920	39.13±0.2182	42.86±0.3012
SSIM	Institution I	Institution II	Institution III	Institution IV
FBP	0.8316±0.03167	0.7563±0.03559	0.7936±0.02883	0.7152±0.02605
FedAvg	0.8641±0.02679	0.8933±0.02254	0.9017±0.02387	0.9063±0.01998
Single_institution	0.9864±0.00763	0.9837±0.00463	0.9897±0.00627	0.9752±0.00798
Multi_institutions	0.9867±0.00323	0.9826±0.00485	0.9763±0.00935	0.9747±0.00845
SC-FL	0.9726±0.00988	0.9779±0.00818	0.9766±0.00900	0.9722±0.00960
icP2P-FL	0.9833±0.00881	0.9790±0.00761	0.9737±0.00852	0.9781±0.00886

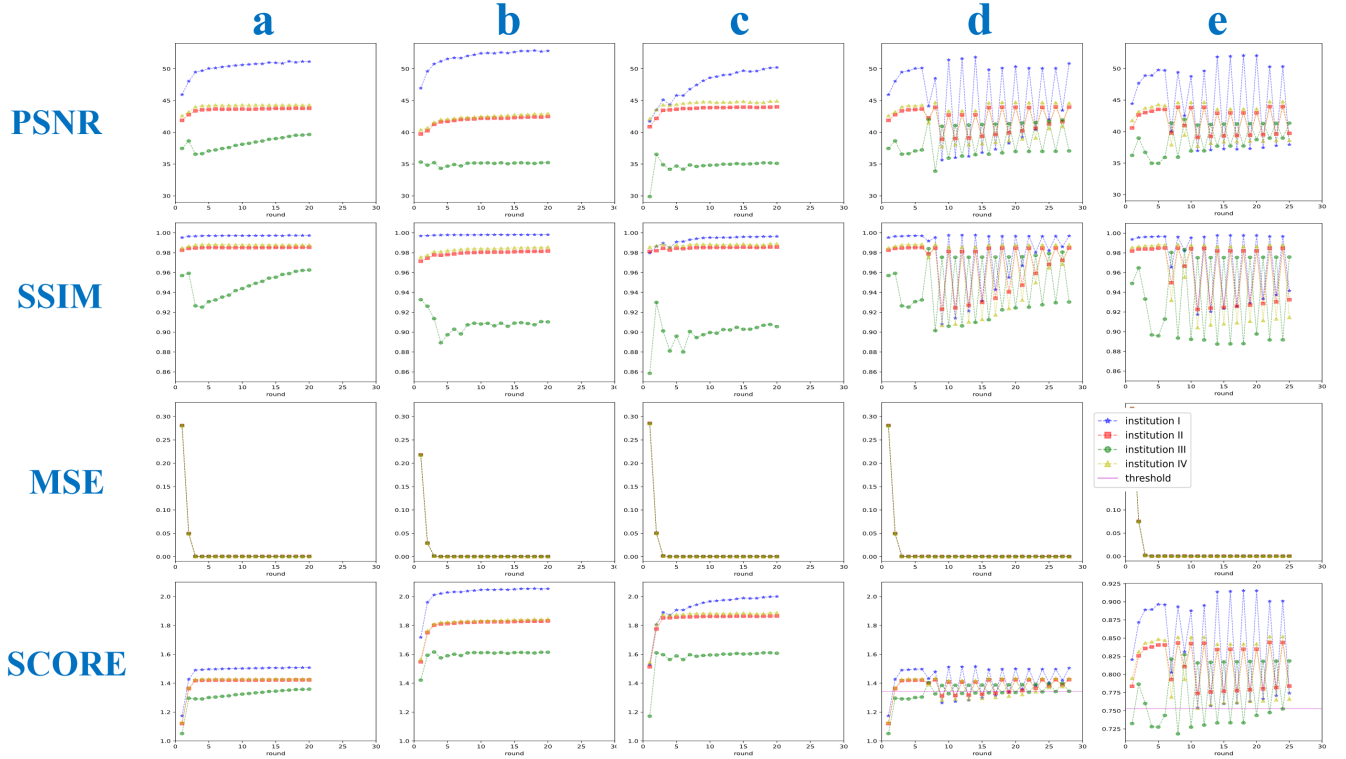


Figure 4: Comparison of quantitative indicators of icP2P-FL under different conditions. (a) Training sequence: [Institution I, Inst. II, Inst. III, Inst. IV], without intermediate controller; (b) Training sequence: [Institution IV, Inst. III, Inst. II, Inst. I], without intermediate controller; (c) Training sequence: [Institution III, Inst. II, Inst. I, Inst. IV], without intermediate controller; (d) Training sequence: [Institution I, Inst. II, Inst. III, Inst. IV], with intermediate controller, determination threshold is 1.343; (e) Training sequence: [Institution I, Inst. II, Inst. III, Inst. IV], with intermediate controller, adjusted the proportion of PSNR and SSIM on the Performance Assessment Module, and the determination threshold is 0.752

artifacts as indicated by the red and blue arrows due to the normal L1 loss function in the SC-FL.

The proposed icP2P-FL method provides a better denoising effect of image reconstruction, and obtains similar denoising and generalization performance visually with the CL-MI method, as

shown in the zoomed-in ROIs. The main reason could be that the continuous learning strategy with an intermediate controller for training tuning in the proposed icP2P-FL allows each local model to perform personalized denoising tasks for image reconstruction, i.e., the CT data acquired with different protocols from different protocols scanners, and reduces the side effects of knowledge forgetting. Moreover, the quantitative assessments of Table 1 indicate that the proposed icP2P-FL method obtains superior performance which is similar to the CL-MI method, highlighting that the proposed icP2P-CL reaches an accuracy similar with the CL-MI method trained with centralized data from all the institutions.

3.3 Ablation experiments

The inclusion of the intermediate controller is crucial within the icP2P-FL framework. To assess its significance, we conducted ablation experiments under various scenarios. These experiments involved the removal of the intermediate controller, the training queue order adjustment strategy, and the training round number adjustment strategy. Figure 4 shows the performance variation of the model tested on the characteristic dataset for different experiments. In specific experiments without the adaptive adjustment of institutional training order, it is observed that each fixed order tends to converge towards a relatively stable value in each evaluation metric. Furthermore, different order scenarios result in distinct final values. When focusing on Institution II results of Figure 5 as the subject of observation, it becomes apparent that the inclusion of intermediate controller can enhance the performance of Institution II under certain circumstances. Consequently, by incorporating the intermediate controller and adjusting the training sequence and number of training iterations for Institution II, the model's overall training becomes more inclined towards Institution II. Simultaneously, this approach ensures that the training effects of other institutions remain within an acceptable range.

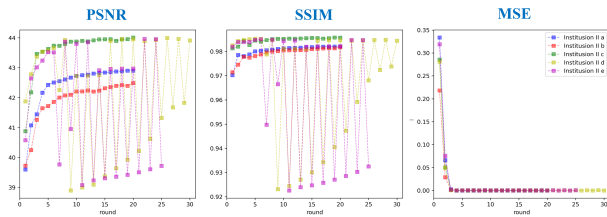


Figure 5: Comparison of quantitative indicators of Institution II under different conditions.

4 Conclusion

This work proposes icP2P-FL as a novel P2P federated learning framework for training personally local model for CT image denoising from different institutions. The motivation behind the proposed icP2P-FL is to enable efficient and effective denoising of CT images with large heterogeneity directly from different protocols and different scanners. The experimental results show that the proposed icP2P-FL method can achieve better performance for robust cross-institution CT denoising for image reconstruction than the existing methods, and reaches an accuracy similar to a model trained

with centralized data from all the institutions. In future studies, additional real clinical datasets should be incorporated to further validate its performance.

Acknowledgment

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References

- [1] 2021. Low Dose CT Grand Challenge. <https://www.aapm.org/GrandChallenge/LowDoseCT/>
- [2] Matthias De Lange, Rahaf Aljundi, Marc Masana, Sarah Parisot, Xu Jia, Aleš Leonardis, Gregory Slabaugh, and Tinne Tuytelaars. 2021. A continual learning survey: Defying forgetting in classification tasks. *IEEE transactions on pattern analysis and machine intelligence* 44, 7 (2021), 3366–3385.
- [3] Jiahua Dong, Lixu Wang, Zhen Fang, Gan Sun, Shichao Xu, Xiao Wang, and Qi Zhu. 2022. Federated class-incremental learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10164–10173.
- [4] Gokberk Elmas, Salman UH Dar, Yilmaz Korkmaz, Emir Ceyani, Burak Susam, Muzaffer Ozbey, Salman Avestimehr, and Tolga Çukur. 2022. Federated learning of generative image priors for MRI reconstruction. *IEEE Transactions on Medical Imaging* (2022).
- [5] Chun-Mei Feng, Yunlu Yan, Shanshan Wang, Yong Xu, Ling Shao, and Huazhu Fu. 2022. Specificity-preserving federated learning for MR image reconstruction. *IEEE Transactions on Medical Imaging* (2022).
- [6] Ji He and Jianhua Ma. 2019. Radon inversion via deep learning. In *Medical Imaging 2019: Physics of Medical Imaging*, Vol. 10948. SPIE, 244–249.
- [7] Ji He, Yan Yang, Yongbo Wang, Dong Zeng, Zhaoying Bian, Hao Zhang, Jian Sun, Zongben Xu, and Jianhua Ma. 2018. Optimizing a parameterized plug-and-play ADMM for iterative low-dose CT reconstruction. *IEEE transactions on medical imaging* 38, 2 (2018), 371–382.
- [8] Timothée Lesort, Vincenzo Lomonaco, Andrei Stoian, Davide Maltoni, David Filliat, and Natalia Diaz-Rodriguez. 2020. Continual learning for robotics: Definition, framework, learning strategies, opportunities and challenges. *Information fusion* 58 (2020), 52–68.
- [9] Danyang Li, Hao Wang, Jingyi Liao, Mingqiang Meng, Cuidie Zeng, Yuting Wang, Dong Zeng, and Jianhua Ma Sr. 2023. Semi-centralized federated learning network for low-dose CT imaging. In *Medical Imaging 2023: Physics of Medical Imaging*, Vol. 12463. SPIE, 1024–1028.
- [10] Wei Liu, Li Chen, and Wenyi Zhang. 2022. Decentralized federated learning: Balancing communication and computing costs. *IEEE Transactions on Signal and Information Processing over Networks* 8 (2022), 131–143.
- [11] Michael McCloskey and Neal J Cohen. 1989. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of learning and motivation*. Vol. 24. Elsevier, 109–165.
- [12] Cynthia H McCollough, Lifeng Yu, James M Kofler, Shuai Leng, Yi Zhang, Zhoubo Li, and Rickey E Carter. 2015. Degradation of CT low-contrast spatial resolution due to the use of iterative reconstruction and reduced dose levels. *Radiology* 276, 2 (2015), 499–506.
- [13] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*. PMLR, 1273–1282.
- [14] III DATA COVERED BY THIS POLICY, IVDNOTCB THIS, and POLICY V BENEFITS OF RELEASING OR. 2003. CDC/ATSDR Policy on Releasing and Sharing Data. (2003).
- [15] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*. Springer, 234–241.
- [16] Micah J Sheller, G Anthony Reina, Brandon Edwards, Jason Martin, and Spyridon Bakas. 2019. Multi-institutional deep learning modeling without sharing patient data: A feasibility study on brain tumor segmentation. In *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: 4th International Workshop, BrainLes 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 16, 2018, Revised Selected Papers, Part I 4*. Springer, 92–104.
- [17] Tobias Wink and Zoltan Nochta. 2021. An approach for peer-to-peer federated learning. In *2021 51st Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*. IEEE, 150–157.
- [18] Chuhan Wu, Fangzhao Wu, Lingjuan Lyu, Yongfeng Huang, and Xing Xie. 2022. Communication-efficient federated learning via knowledge distillation. *Nature communications* 13, 1 (2022), 2032.
- [19] Dufan Wu, Kyungsang Kim, Georges El Fakhri, and Quanzheng Li. 2017. Iterative low-dose CT reconstruction with priors trained by artificial neural network. *IEEE*

- transactions on medical imaging* 36, 12 (2017), 2479–2486.
- [20] Jie Xu, Benjamin S Glicksberg, Chang Su, Peter Walker, Jiang Bian, and Fei Wang. 2021. Federated learning for healthcare informatics. *Journal of Healthcare Informatics Research* 5 (2021), 1–19.
 - [21] Ziyuan Yang, Wenjun Xia, Zexin Lu, Yingyu Chen, Xiaoxiao Li, and Yi Zhang. 2022. Hypernetwork-based personalized federated learning for multi-institutional CT imaging. *arXiv preprint arXiv:2206.03709* (2022).
 - [22] Syed Waqas Zamir, Aditya Arora, Salman Khan, Munawar Hayat, Fahad Shah-baz Khan, and Ming-Hsuan Yang. 2022. Restormer: Efficient transformer for high-resolution image restoration. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 5728–5739.
 - [23] Dong Zeng, Jing Huang, Zhaoying Bian, Shanzhou Niu, Hua Zhang, Qianjin Feng, Zhengrong Liang, and Jianhua Ma. 2015. A simple low-dose x-ray CT simulation from high-dose scan. *IEEE transactions on nuclear science* 62, 5 (2015), 2226–2233.
 - [24] Zhou, Bo and Miao, Tianshun and Mirian, Niloufar and Chen, Xiongchao and Xie, Huidong and Feng, Zhicheng and Guo, Xueqi and Li, Xiaoxiao and Zhou, S. Kevin and Duncan, James S. and Liu, Chi. 2023. Federated Transfer Learning for Low-Dose PET Denoising: A Pilot Study With Simulated Heterogeneous Data. *IEEE Transactions on Radiation and Plasma Medical Sciences* 7, 3 (2023), 284–295. doi:10.1109/TRPMS.2022.3194408
 - [25] Bo Zhu, Jeremiah Z Liu, Stephen F Cauley, Bruce R Rosen, and Matthew S Rosen. 2018. Image reconstruction by domain-transform manifold learning. *Nature* 555, 7697 (2018), 487–492.

Continual Learning Strategies for Personalized Mental Well-being Monitoring from Mobile Sensing Data

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Abstract

Automatic prediction of psychological stress using mobile sensing data is critical for developing personalized mental health monitoring tools. Traditional approaches typically train static models offline, limiting their applicability in real-world scenarios characterized by evolving user behaviors and data streams. To address these challenges, we formulate stress prediction as a continual learning (CL) regression task, where each user constitutes a distinct sequential task. Within this framework, we evaluate two adaptation strategies: (1) a replay-based approach, leveraging recurrent neural networks combined with an episodic memory buffer; and (2) a prompt-based method, where task-specific prompts are jointly trained with a transformer backbone and combined with the memory buffer to enable task-aware adaptation across users. Experiments conducted on a filtered subset of the StudentLife dataset, comprising multimodal behavioral and contextual data, demonstrate that prompt-based methods achieve competitive predictive performance. Our results underline the potential of continual learning paradigms to enable scalable and adaptive stress prediction systems.

CCS Concepts

• **Computing methodologies** → **Lifelong machine learning; Machine learning; Artificial intelligence**; • **Computer systems organization** → **Neural networks**.

Keywords

stress prediction, continual learning, mobile sensing, prompt tuning

ACM Reference Format:

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1 Introduction

Psychological stress is increasingly recognized as a major health concern, particularly among university students, whose daily lives are characterized by intense academic demands, complex social interactions, and significant lifestyle changes [3, 22, 24]. Chronic exposure to high levels of stress has been linked to a broad spectrum of adverse health outcomes, including anxiety disorders, depression, impaired cognitive performance, sleep disturbances, and reduced physical health [7, 32, 34]. Consequently, there is a growing interest in the scientific community in developing automated, reliable methods to continuously monitor and predict stress levels in daily life.

Recent advances in wearable and smartphone technology have provided unprecedented opportunities for real-time and unobtrusive monitoring of individuals' psychological and behavioral states [12, 28]. These devices collect a wealth of multimodal data, including physiological signals (such as heart rate variability and galvanic skin response), behavioral patterns (such as physical activity and smartphone usage), and contextual information (such as location, ambient noise, and social interactions) [17, 36]. Leveraging this data, affective computing and mobile sensing research have made significant strides in creating predictive models of psychological stress that are applicable to everyday settings. Despite the progress achieved thus far, existing approaches primarily rely on static machine learning models trained offline using aggregated data from multiple individuals [2, 10, 28]. However, these methods struggle to generalize in real-world scenarios, as they are unable to adapt to temporal shifts in user behavior and require full retraining when exposed to new users or changing data streams.

In response to these challenges, this paper introduces a novel paradigm by formulating the stress prediction problem within a continual learning (CL) framework. Continual learning is specifically designed to tackle scenarios where data distributions are non-stationary and tasks evolve incrementally over time. By framing each user as a distinct incremental learning task, we create a model capable of dynamically adapting to new individuals while maintaining performance on previously learned users, thus inherently supporting personalized adaptation and sustained learning over extended periods.

Within this continual learning setting, we explore and compare two major approaches. The first is a memory replay-based approach, leveraging recurrent neural network architectures, such as standard RNNs and bidirectional long short-term memory (BiLSTM), combined with episodic memory buffers to mitigate catastrophic forgetting of previously learned tasks. The second is an extension

of the former, where prompt-based mechanisms are integrated into the memory replay framework to further enhance knowledge retention across tasks. In this scenario, both the transformer backbone and the task-specific prompts are continuously updated, with the prompts being refined using replayed data from previous tasks to support continual adaptation.

To rigorously evaluate the proposed approaches, we utilize a carefully curated subset of the StudentLife dataset [36]. This dataset comprises comprehensive smartphone-sensed behavioral data and detailed self-reported stress annotations from a population of university students collected over an extended observation period. The dataset provides rich multimodal signals, enabling a robust evaluation of stress prediction in realistic conditions.

Our experimental results demonstrate that the prompt-based approach achieves comparable or superior performance compared to standard memory replay methods, highlighting the benefit of incorporating task-aware adaptation.

Our main contributions are as follows:

- We propose a novel continual learning formulation of the stress prediction problem, tailored for dynamic real-world scenarios.
- We compare two continual learning strategies: memory replay using RNN-based models and prompt-based adaptation using transformer models.
- We empirically validate the effectiveness of behavioral and contextual features extracted from smartphone data for psychological stress prediction in both task- and domain-incremental learning settings.

Through these contributions, our work contributes to the development of stress prediction systems that are better suited for real-world deployment, supporting progress in affective computing and digital health.

2 Related Work

The automatic estimation of psychological stress has been widely investigated in the field of affective computing, leveraging signals from mobile and wearable devices to infer mental states in everyday life [6, 12].

Designed to study stress in daily life, the StudentLife dataset [36] consists of passive sensing and self-report data collected from 48 university students over a 10-week period. The data revealed that increased academic demands were associated with reduced physical activity, shorter sleep, and lower positive affect.

Motivated by these findings, numerous studies have proposed predictive models using both traditional and deep learning approaches. Classical methods such as logistic regression [28, 36] and decision trees [2] remain popular due to their interpretability and robustness in data-scarce conditions. Muaremi et al. [28] classified workplace stress using smartphone-derived features such as communication logs and heart rate variability (HRV), while Wang et al. [36] demonstrated associations between smartphone behavior, self-reported stress, and academic performance.

In parallel, machine learning models have sought to identify the most informative behavioral signals for stress prediction. DaSilva et al. [10] used penalized regression identifying social activity, location variance, and ambient noise as significant stress correlates.

Bonafonte et al. [4] systematically compared feature subsets (e.g., WiFi, motion, phone usage), and found that mobility and device-logging patterns (e.g., night-time movement, battery events) were more predictive than communication metrics such as call or SMS frequency. These findings suggest that physical activity and rest patterns may serve as more reliable indicators of stress than communication behavior alone.

To support these modeling efforts, a variety of datasets have been developed across both controlled and real-world environments. In laboratory settings, the WESAD dataset [31] includes physiological and motion signals collected from 15 participants under three affective conditions: neutral, stress, and amusement. In contrast, real-world studies such as those by Muaremi et al. [28] and K-EmoPhone [17] provide multimodal smartphone and wearable recordings over multiple days, annotated with self-reported stress and affective states. Larger-scale longitudinal studies such as TILES [15] and Tesseract [27] further extend this direction by integrating wearable, mobile, and self-report data in workplace environments, enabling the study of stress over extended periods.

Despite these advances, most existing models rely on static training-test splits and assume a fixed population, limiting their ability to adapt to new users or evolving behavioral patterns. As Lazarou et al. [21] emphasize, the widespread adoption of wearable technologies enables continuous stress monitoring, yet current models often lack mechanisms for incremental adaptation and long-term personalization.

A central challenge in this context is the heterogeneity of behavioral and physiological signals across users, which has led to increasing interest in personalized or user-aware models. Shaw et al. [33] and Luo et al. [26] introduced deep multitask architectures with shared and user-specific components, allowing better adaptation to individual variability. In particular, their designs enable dynamic grouping of users based on latent similarity, which facilitates parameter sharing and mitigates the cold-start problem. Although not explicitly framed as continual learning, these approaches align with task-based continual learning paradigms, in which each user or user cluster can be considered a distinct task introduced sequentially.

This perspective resonates with the principles of personalized continual learning, in which models are updated incrementally as data from new users becomes available, without retraining from scratch.

Continual learning in health and affective computing. Continual learning (CL) is an increasingly attractive paradigm for addressing non-stationary data distributions and evolving task structures in real-world machine learning deployments [11, 29]. CL enables models to acquire new information over time while retaining previously learned knowledge, a balance that is especially critical in health-related and affective computing domains characterized by longitudinal data, inter-subject variability, and the need for sustained personalization [30].

In the medical domain, continual learning has been applied to diagnostic prediction, physiological signal interpretation, and medical image segmentation. Gonzalez et al. [13] proposed Lifelong nnU-Net, a continual learning variant of the widely-used nnU-Net,

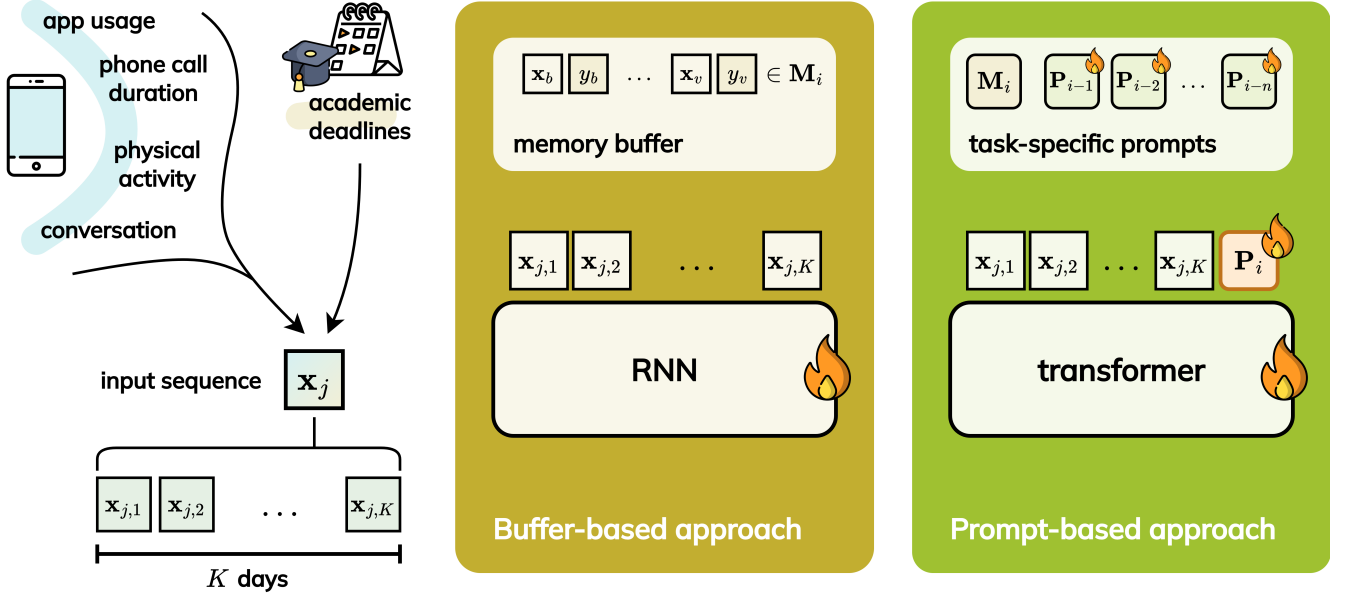


Figure 1: Overview of the proposed method. On the left, the input sequence $\mathbf{x}_j$, collected over the K days preceding a given stress report y_i , consists of behavioral and contextual features such as app usage, phone call duration, physical activity, conversation, and academic deadlines. In the middle, the replay-based approach combines a sequence model, such as an RNN, with a memory buffer of past task examples. On the right, the prompt-based method leverages a transformer encoder and a memory buffer, using learnable task-specific prompts P_i prepended to the input sequence to enable task adaptation.

enabling models to adapt to novel anatomical structures and imaging protocols without catastrophic forgetting. Kiyasseh et al. [20] introduced CLOPS, a framework for biosignal analysis that leverages experience replay and task-specific memory buffers to retain key temporal patterns from streaming physiological data such as ECG and PPG. These methods often integrate regularization strategies (e.g., EWC [19], LwF [25]) or modular architectures to support scalable and stable learning over time [11].

In affective computing, the need for personalization is especially acute due to high inter-subject variability and the dynamic nature of emotional states. Ahmad et al. [1] recently introduced SSOCL, a bi-level self-supervised framework for EEG-based emotion recognition that uses a dynamic buffer and pseudo-labeling to handle continuously streaming and unlabeled physiological data. In parallel, few-shot learning techniques have been effectively employed to address cross-subject variability in EEG emotion recognition, where models pretrained offline can be rapidly fine-tuned to new users with minimal data, as demonstrated in several studies of prototype or metric-learning architectures [39].

Our work frames stress prediction as a continual learning problem by treating each user as an independent task; thereby introducing temporal continuity and lifelong adaptation, and aligning more closely with the dynamic and evolving nature of real-world deployment scenarios.

3 Method

In this section, we describe the methodology adopted to predict students' stress levels through a continual learning framework. Our approach formulates stress estimation as a regression task based on mobile sensing data. An overview of the proposed method is presented in Fig. 1.

3.1 Problem formulation

We pose our investigation as a continual learning regression problem. Given a sequence of T distinct tasks, $\mathcal{T} = \{\tau_1, \dots, \tau_T\}$, each task τ_i is associated with a dataset $\mathcal{D}_i = \{(\mathbf{x}_j, y_j)\}$, where $\mathbf{x}_j$ is an input sample and y_j is its corresponding target value.

Each label y_j represents the stress level self-reported by the student on a given day, and the input sample $\mathbf{x}_j$ is the sequence of sensing data $\{\mathbf{x}_{j,1}, \mathbf{x}_{j,2}, \dots, \mathbf{x}_{j,K}\}$ collected over the K days preceding that report.

We operate in both a domain-incremental learning setting, where each task τ_i corresponds to a specific subset of students. While the input distribution changes across tasks, the output space remains fixed, with stress levels ranging in a shared label space $\mathcal{Y} = \{1, 2, 3, 4, 5\}$, where higher values indicate higher perceived stress.

The objective is to learn a function $f: \mathcal{X} \rightarrow \mathcal{Y}$, parameterized by θ , that generalizes across tasks, while at each task τ_i the model is exposed only to data from the corresponding distribution $\mathcal{D}_i$, i.e., $(\mathbf{x}, y) \sim \mathcal{D}_i$.

Our methodology draws inspiration from two main paradigms in continual learning: memory replay (*rehearsal*) with task-agnostic models [5, 8, 9], and recent developments in prompt-based adaptation for transformers [37, 38].

3.2 Buffer-based approach

To reduce forgetting from previous tasks, buffer-based methods store a limited number of samples from previous tasks in a *memory buffer* $\mathbf{M}$. The model update can be summarized as:

$$\langle f, \theta_{i-1}, \mathbf{M}_{i-1} \rangle \xrightarrow{\mathcal{D}_i} \langle f, \theta_i, \mathbf{M}_i \rangle, \quad (1)$$

where θ_i and $\mathbf{M}_i$ denote the model parameters and the memory buffer at the end of task τ_i , respectively.

The training objective is to minimize a regression loss over the sequence of tasks, aiming to preserve performance on both current and previous tasks:

$$\begin{aligned} \arg \min_{\theta_T} \alpha \sum_{i=1}^T \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_i} \left[\mathcal{L}(f(\mathbf{x}; \theta_T), y) \right] \\ + \beta \sum_{i=1}^T \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{M}_i} \left[\mathcal{L}(f(\mathbf{x}; \theta_T), y) \right] \end{aligned} \quad (2)$$

where $\mathcal{L}$ is a regression loss function (e.g., Mean Squared Error), and α and β are weighting coefficients that balance the contributions of the two loss components.

3.3 Prompt-based approach

Recently, transformers have been increasingly adopted in continual learning research, thanks to the capability to condition their behavior through learnable prompt tuning: in this setting, task adaptation is achieved by tuning small *task-specific prompt vectors* while keeping a pre-trained backbone frozen. Since pre-training is not applicable in our setting due to the nature of the data, we instead jointly train the backbone and the prompts across tasks, with a supporting memory buffer to ensure alignment between prompts learned in older tasks and the continually-updated backbone. In this way, we are able to extend the replay-based approach defined in the previous section with a strong and standardized conditioning mechanism.

In this formulation, a transformer $\mathcal{T}$ is employed as the backbone encoder, projecting the sequence of sensing features into a d -dimensional representation. For each task τ_i , the input sequence $\mathbf{x}_j$ is augmented by prepending a set of N learnable task-specific prompts, $P_i = \{p_1^i, \dots, p_N^i\}$. The resulting input to the transformer is defined as:

$$\mathcal{T}(P_i \parallel \mathbf{x}_j), \quad (3)$$

where $\parallel$ denotes concatenation along the temporal axis.

During training, both the backbone transformer $\mathcal{T}$ and the parameters of each prompt P_i are updated. As mentioned above, to ensure that the prompts $\{P_1, \dots, P_{i-1}\}$ associated with previous tasks $\{\tau_1, \dots, \tau_{i-1}\}$ remain well-adapted, a memory buffer $\mathbf{M}_i$ is employed to store samples from past tasks, which are replayed during training on task τ_i to refine the corresponding prompts. At inference time, the task identity is assumed to be known a priori, allowing the model to use the corresponding learned prompt set.

The learning objective remains the same as in Equation 2, with the optimization extended to include the prompt parameters.

4 Experimental evaluation

In this section, we present the experimental evaluation of the proposed method. Section 4.1 presents the dataset used in our experiments and explains how it was preprocessed. The employed architectures are detailed in Section 4.2, followed by the evaluation metrics in Section 4.3. The training procedure and the experimental results are presented in Section 4.4 and Section 4.5, respectively.

4.1 Dataset

We used the StudentLife dataset [36], which includes multimodal data from 48 undergraduate and graduate students at Dartmouth College over a 10-week spring term. The dataset contains over 53 GB of continuous sensor data, 32,000 self-reports, and several pre/post-study surveys covering mental health, lifestyle factors, and academic-related attitudes.

Given that self-reported stress is the target variable in our study, we focused on the corresponding questionnaire data provided by each participant. However, as not all participants contributed responses evenly across the 10-week period, we applied a filtering step to ensure both temporal consistency and sufficient response density. We retained only those who completed at least 30 stress questionnaires within a time window of 60 to 75 days, resulting in a final subset of 16 students. Additionally, we excluded one outlier who submitted an high number of responses, many of which extended beyond the expected 10-week timeframe. Figure 2 reports the number of days with available stress self-report data for each participant, while Fig 3 reports the number of stress responses for the selected subset.

Data processing. Among the various data types available in the StudentLife dataset, we selected a limited subset of features consisting of *phone call duration*, *conversation*, *physical activity*, *app usage* and *academic deadlines*.

To streamline the analysis and mitigate the risk of bias or information leakage, we excluded features deemed redundant or less informative (e.g., additional phone data, location information, and seating logs). Instead, we focused on behavioral and contextual signals while avoiding variables that may directly reveal or strongly correlate with the target outcome.

The selected features were extracted and aggregated per day, as described below:

- **Phone call duration** includes both the total time spent on completed incoming and outgoing calls, expressed in minutes, and the number of such interactions. Missed or rejected calls are excluded, as they do not reflect active engagement.
- **Conversation** consists of time periods inferred from microphone data via acoustic analysis as indicative of nearby spoken interactions. Both the number of detected periods and their total duration in minutes were extracted.
- **Physical activity** is derived from smartphone sensor data using a classifier that operates continuously with duty cycling to optimize energy consumption. The resulting activity is classified into three categories: stationary, walking, and

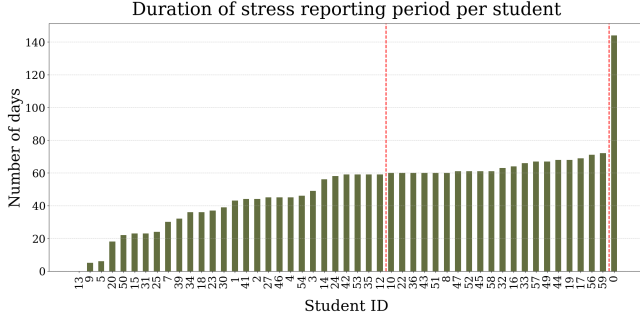


Figure 2: Length of the stress self-reporting period (in days) for each student, based on the interval between their first and last submission.

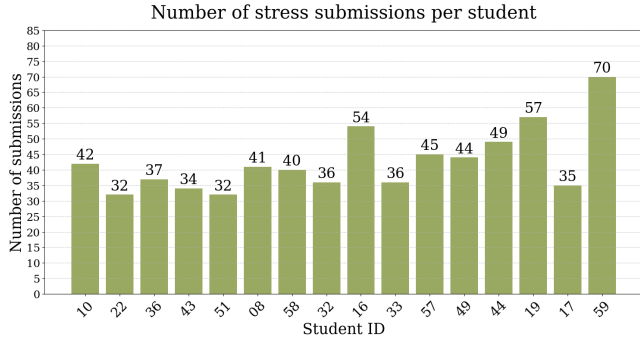


Figure 3: Number of stress self-report submissions for each student in the filtered subset.

running. The total daily duration in minutes was computed for each activity state.

- **App usage** captures the set of mobile applications accessed by the participant. To organize this data meaningfully, a clustering approach was applied using BART [23], a transformer-based model capable of generating semantically rich representations. Each app was assigned to one of ten predefined semantic categories: “social media and communication”, “productivity and business”, “entertainment and media”, “health and fitness”, “finance and shopping”, “utilities and tools”, “games”, “education”, “travel and navigation”, and “photography”. Based on this categorization, two features were computed for each day: the overall time spent using the phone, and the usage time in minutes for each category.
- **Academic deadlines** represent the daily number of academic events scheduled for each participant, including assignments, quizzes, projects, and exams. These data were obtained from a structured log of time-stamped academic events.

Sensor-based features, such as *conversation*, may suffer from occasional missing data due to sensor unavailability, battery optimization mechanisms, or interruptions in background processes. To ensure a consistent temporal resolution, we applied linear interpolation to estimate missing values on a per-day basis.

Stress levels, ranging from 1 to 5, were obtained from participants’ self-reports, with higher values indicating greater perceived stress. To predict the stress level on a given day, the input consists of a sequence of K feature vectors, each representing the full set of the aforementioned daily variables over the K preceding days.

4.2 Architectures

To investigate the effectiveness of the proposed approaches, we employed the following sequence modeling architectures:

- **Recurrent Neural Network (RNN)** [16]: a two-layer unidirectional vanilla RNN with a hidden size of 64 and a dropout rate of 0.3. The model processes input sequences one timestep at a time, maintaining a hidden state to capture temporal dependencies. Due to its simple recurrent structure, it is susceptible to vanishing gradients when modeling long-range dependencies. We train this model following the buffer-based approach described in Sect. 3.2.
- **Bidirectional Long Short-Term Memory (BiLSTM)** [14]: a two-layer bidirectional LSTM with a hidden size of 96 per direction and a dropout rate of 0.3. Unlike the vanilla RNN, the LSTM incorporates gating mechanisms and a dedicated cell state to preserve information over longer temporal windows. The bidirectional configuration enables the model to leverage both past and future context at each timestep, enhancing its ability to encode structured sequential data. We train this model following the buffer-based approach described in Sect. 3.2.
- **Transformer encoder** [35]: a two-layer transformer encoder, with each layer comprising four attention heads and an embedding dimension of 256. A final linear layer follows the encoder, projecting the internal representations into a single output dimension. To preserve the temporal structure of the input, positional encodings are added to the input embeddings. We train this model following the prompt-based approach described in Sect. 3.3.

4.3 Metrics

To evaluate performance, we use two standard regression metrics: Mean Squared Error (MSE) and Mean Absolute Error (MAE). For each metric, we provide the mean and standard deviation on the test split defined by the employed datasets, over three runs for all experiments.

Note that, while for the recurrent architectures this evaluation setting is consistent with a pure domain-incremental scenario, the transformer encoder operates in a joint domain- and task-incremental setting, as prompt selection at inference time depends on task identity.

4.4 Training procedure

We split the available data into training, validation, and test sets using a 70%-10%-20% ratio, ensuring that each student’s data is evenly distributed across the splits.

Training is carried out for 50 epochs per task using the Adam optimizer [18], with a learning rate of 0.001 and a batch size of 32, setting $\alpha = 1$ and $\beta = 0.5$ for the training objective defined in Eq. 2.

Table 1: Comparison of buffer-only and prompt-augmented approaches for stress prediction across 16 tasks, with one student per task. Results are reported by varying the memory buffer size ($|M|$). In the prompt-based setting, we also vary the number of learned prompts per task ($|P_i|$). Performance is reported in terms of MSE ($\downarrow$) and MAE ($\downarrow$), with standard deviation.

	16 tasks: one student per task					
	$ M =20$		$ M =50$		$ M =100$	
	MSE	MAE	MSE	MAE	MSE	MAE
RNN [16]	2.644 ± 0.310	1.109 ± 0.015	2.543 ± 0.383	1.096 ± 0.016	2.703 ± 0.169	1.107 ± 0.008
BiLSTM [14]	2.657 ± 0.103	1.116 ± 0.007	2.705 ± 0.012	1.109 ± 0.005	2.696 ± 0.038	1.110 ± 0.005
Transformer w/ $ P_i =1$	2.054 ± 0.003	1.083 ± 0.001	2.046 ± 0.006	1.051 ± 0.006	2.045 ± 0.003	1.014 ± 0.001
Transformer w/ $ P_i =2$	2.157 ± 0.171	1.086 ± 0.001	2.052 ± 0.010	1.084 ± 0.001	2.029 ± 0.011	1.069 ± 0.012
Transformer w/ $ P_i =4$	2.938 ± 0.051	1.269 ± 0.031	2.596 ± 0.461	1.158 ± 0.001	2.318 ± 0.084	1.103 ± 0.022

Table 2: Comparison of buffer-only and prompt-augmented approaches for stress prediction across 8 tasks, with two students per task. Results are reported by varying the memory buffer size ($|M|$). In the prompt-based setting, we also vary the number of learned prompts per task ($|P_i|$). Performance is reported in terms of MSE ($\downarrow$) and MAE ($\downarrow$), with standard deviation.

	8 tasks: two students per task					
	$ M =20$		$ M =50$		$ M =100$	
	MSE	MAE	MSE	MAE	MSE	MAE
RNN [16]	2.202 ± 0.244	0.963 ± 0.026	2.093 ± 0.209	0.987 ± 0.045	2.381 ± 0.054	0.976 ± 0.018
BiLSTM [14]	2.298 ± 0.078	1.004 ± 0.029	2.289 ± 0.022	0.985 ± 0.004	2.245 ± 0.017	0.978 ± 0.003
Transformer w/ $ P_i =1$	1.707 ± 0.018	0.980 ± 0.001	1.762 ± 0.106	0.976 ± 0.008	1.688 ± 0.021	0.934 ± 0.032
Transformer w/ $ P_i =2$	1.730 ± 0.050	0.976 ± 0.009	1.719 ± 0.039	0.952 ± 0.039	1.700 ± 0.004	0.932 ± 0.013
Transformer w/ $ P_i =4$	1.948 ± 0.050	0.964 ± 0.010	1.929 ± 0.120	0.963 ± 0.006	1.915 ± 0.032	0.948 ± 0.016

Table 3: Comparison of buffer-only and prompt-augmented approaches for stress prediction across 4 tasks, with four students per task. Results are reported by varying the memory buffer size ($|M|$). In the prompt-based setting, we also vary the number of learned prompts per task ($|P_i|$). Performance is reported in terms of MSE ($\downarrow$) and MAE ($\downarrow$), with standard deviation.

	4 tasks: four students per task					
	$ M =20$		$ M =50$		$ M =100$	
	MSE	MAE	MSE	MAE	MSE	MAE
RNN [16]	1.462 ± 0.196	0.968 ± 0.039	2.236 ± 0.269	1.097 ± 0.061	2.431 ± 0.095	1.091 ± 0.038
BiLSTM [14]	2.048 ± 0.031	1.040 ± 0.005	2.042 ± 0.058	1.042 ± 0.017	2.047 ± 0.045	1.044 ± 0.014
Transformer w/ $ P_i =1$	1.565 ± 0.069	0.948 ± 0.001	1.529 ± 0.015	0.945 ± 0.004	1.354 ± 0.052	0.930 ± 0.015
Transformer w/ $ P_i =2$	1.540 ± 0.023	0.943 ± 0.004	1.512 ± 0.001	0.933 ± 0.006	1.499 ± 0.055	0.927 ± 0.019
Transformer w/ $ P_i =4$	1.634 ± 0.103	0.941 ± 0.003	1.581 ± 0.025	0.920 ± 0.007	1.513 ± 0.031	0.900 ± 0.017

All hyperparameters were selected based on performance on the validation set.

Given the limited number of samples per task, we employ relatively small buffer sizes $|M|$, experimenting with buffers of 20, 50, and 100 samples.

4.5 Results

In this section, we present the results of our stress prediction experiments, comparing buffer-based methods with a prompt-augmented variant. All models are evaluated using the performance metrics defined in Sec. 4.3.

To establish meaningful baselines for our continual learning methods, Tab. 4 reports results obtained by training each model on the full dataset, using the architectures described in Sec. 4.2.

Table 4: Performance of joint training across different model architectures. Comparison of RNN, BiLSTM, and transformer with varying prompt set size $|P|$. Performance is reported in terms of MSE ($\downarrow$) and MAE ($\downarrow$), including standard deviations.

Joint training		
	MSE	MAE
RNN [16]	1.905 ± 0.081	1.131 ± 0.040
BiLSTM [14]	1.834 ± 0.055	1.097 ± 0.021
Transformer w/ $ P = 4$	1.866 ± 0.030	1.021 ± 0.007
Transformer w/ $ P = 8$	1.077 ± 0.032	0.844 ± 0.012
Transformer w/ $ P = 16$	1.161 ± 0.035	0.778 ± 0.014
Transformer w/ $ P = 32$	1.159 ± 0.017	0.827 ± 0.013
Transformer w/ $ P = 64$	1.235 ± 0.085	0.919 ± 0.040

Table 5: Performance of continual training without rehearsal: a comparison of RNN, BiLSTM, and transformer architectures. Performance is reported in terms of MSE ($\downarrow$) and MAE ($\downarrow$), including standard deviations.

No rehearsal training		
16 tasks: one student per task		
	MSE	MAE
RNN [16]	3.335 ± 0.118	1.251 ± 0.038
BiLSTM [14]	3.045 ± 0.098	1.181 ± 0.013
Transformer w/o prompts	2.185 ± 0.009	1.102 ± 0.004
8 tasks: two students per task		
	MSE	MAE
RNN [16]	2.941 ± 0.100	1.108 ± 0.036
BiLSTM [14]	2.684 ± 0.071	1.086 ± 0.023
Transformer w/o prompts	1.832 ± 0.008	1.021 ± 0.004
4 tasks: four students per task		
	MSE	MAE
RNN [16]	2.639 ± 0.081	1.184 ± 0.026
BiLSTM [14]	2.415 ± 0.074	1.105 ± 0.017
Transformer w/o prompts	1.704 ± 0.010	0.985 ± 0.006

As a lower bound, Tab. 5 shows results from continual learning experiments without any replay buffer. The baseline results indicate that recurrent architectures yield the highest prediction errors, reflecting their limited ability to model long-range task dependencies. Transformer-based encoders consistently outperform recurrent models, achieving lower error rates in both joint and continual settings. In the joint training setup, the best performance is achieved with a prompt set size of $|P| = 8$, while larger sizes provide no further benefit and may even degrade performance, suggesting over-parameterization.

We evaluated our approaches in a continual learning setting by splitting the 16 students into different numbers of tasks. The results for the 16-task, 8-task, and 4-task configurations are presented in Tab. 1, Tab. 2, and Tab. 3, respectively.

Across all tested configurations, transformer-based architectures consistently outperform recurrent models, achieving significantly lower error metrics. Minimal prompt configurations ($|P_i| = 1$ or $|P_i| = 2$) yield the best results, whereas larger prompt sets often degrade performance, indicating potential overfitting or over-parameterization. Introducing even a single prompt per task improves transformer performance across all scenarios compared to their non-prompted counterparts, although this comes at the cost of requiring the task identifier at inference time.

Performance generally benefits from increasing the memory buffer size $|M|$, yet the improvement becomes marginal for very large buffers ($|M| = 100$), suggesting that the proposed methods can achieve competitive results even with reduced memory capacity. This advantage is particularly relevant in real-world stress monitoring applications where storage resources are limited.

Overall, these findings highlight the benefits of combining transformer architectures with prompt-based replay and underscore the importance of carefully tuning the number of learned prompts per task to avoid performance degradation.

5 Discussion

A key limitation identified in our analysis is the tendency of the model to underfit in certain tasks. This limitation primarily arises from the low reliability of the multimodal data collected via smartphones. The stress target exhibits high short-term variability, which constrains the model's ability to learn stable and meaningful patterns. Data quality is further compromised by artifacts such as environmental noise, intermittent signal loss, and inconsistent device usage by participants. As a result, the model struggles to learn robust representations and to generalize beyond the training data. To address these limitations, future work should focus on two key directions: designing more resilient multimodal fusion architectures and collecting higher-quality datasets.

6 Conclusion

In this work, we addressed the challenge of stress prediction in real-world scenarios, aiming to overcome the limitations of static and traditional machine learning models. To mimic realistic distribution shifts, we organized the data into tasks comprising one, two, or four students each. Within this framework, we investigated two adaptation strategies: a memory-based replay method employing recurrent neural networks, and a prompt-based approach leveraging a transformer backbone with task-specific tunable prompts.

Experiments on a subset of the StudentLife dataset demonstrate that, despite the need to store and retrieve task-specific prompts at inference time, prompt-based adaptation achieves lower prediction error than buffer-only replay methods. Optimal performance is obtained with minimal prompt sets, whereas larger configurations may introduce unnecessary complexity. Moreover, performance remains competitive even with reduced memory, underscoring the practicality of the approach for resource-constrained stress monitoring scenarios.

Future work could explore more advanced prompting strategies, methods for reducing or removing the need for explicit task identifiers, and broader applications of the proposed framework across different domains of mental health and human behavior modeling.

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References

- [1] Adnan Ahmad, Bahareh Nakisa, and Mohammad Naim Rastgoo. 2025. Robust Emotion Recognition via Bi-Level Self-Supervised Continual Learning. *arXiv preprint arXiv:2505.10575* (2025).
- [2] Seyed Amir Hossein Aqajari, Sina Labbaf, Phuc Hoang Tran, Brenda Nguyen, Milad Asgari Mehrabadi, Marco Levorato, Nikil Dutt, and Amir M Rahmani. 2023. Context-aware stress monitoring using wearable and mobile technologies in everyday settings. *arXiv preprint arXiv:2401.05367* (2023).
- [3] Saba Asif, Azka Mudassar, Talala Zainab Shahzad, Mobeen Raouf, and Tehmina Pervaiz. 2020. Frequency of depression, anxiety and stress among university students. *Pakistan journal of medical sciences* 36, 5 (2020), 971.
- [4] Irene Bonafonte, Cristina Bustos, Abraham Larrazolo, Gilberto Lorenzo Martínez Luna, Adolfo Guzmán Arenas, Xavier Baró, Isaac Tourgeman, Mercedes Balcells, and Agata Lapedriza. 2023. Analyzing the contribution of different passively collected data to predict Stress and Depression. In *2023 International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*.
- [5] Lucas Caccia, Rahaf Aljundi, Nader Asadi, Tinne Tuytelaars, Joelle Pineau, and Eugene Belilovsky. 2022. New Insights on Reducing Abrupt Representation Change in Online Continual Learning. In *International Conference on Learning Representations*.
- [6] Yekta Said Can, Bert Arnrich, and Cem Ersoy. 2019. Stress detection in daily life scenarios using smart phones and wearable sensors: A survey. *Journal of biomedical informatics* 92 (2019), 103139.
- [7] Adelinda Araújo Candeias, Edgar Galindo, Konrad Reschke, Mariola Bidzan, and Marcus Stueck. 2024. The interplay of stress, health, and well-being: unraveling the psychological and physiological processes. 1471084 pages.
- [8] Arslan Chaudhry, Albert Gordo, Puneet Dokania, Philip Torr, and David Lopez-Paz. 2021. Using hindsight to anchor past knowledge in continual learning. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 35. 6993–7001.
- [9] Arslan Chaudhry, Marcus Rohrbach, Mohamed Elhoseiny, Thalaiyasingam Ajanthan, P Dokania, P Torr, and M Ranzato. 2019. Continual learning with tiny episodic memories. In *Workshop on Multi-Task and Lifelong Reinforcement Learning*.
- [10] Alex W DaSilva, Jeremy F Huckins, Rui Wang, Weichen Wang, Dylan D Wagner, and Andrew T Campbell. 2019. Correlates of stress in the college environment uncovered by the application of penalized generalized estimating equations to mobile sensing data. *JMIR mHealth and uHealth* 7, 3 (2019), e12084.
- [11] Matthias De Lange, Rahaf Aljundi, Marc Masana, Sarah Parisot, Xu Jia, Aleš Leonardis, Gregory Slabaugh, and Tinne Tuytelaars. 2021. A continual learning survey: Defying forgetting in classification tasks. *IEEE transactions on pattern analysis and machine intelligence* 44, 7 (2021), 3366–3385.
- [12] Shruti Gedam and Sanchita Paul. 2021. A review on mental stress detection using wearable sensors and machine learning techniques. *IEEE Access* (2021).
- [13] Camila González, Amin Ranem, Daniel Pinto dos Santos, Ahmed Othman, and Anirban Mukhopadhyay. 2023. Lifelong nnu-net: a framework for standardized medical continual learning. *Scientific Reports* 13, 1 (2023), 9381.
- [14] Alex Graves and Jürgen Schmidhuber. 2005. Framewise phoneme classification with bidirectional LSTM networks. In *Proceedings. 2005 IEEE International Joint Conference on Neural Networks*, 2005, Vol. 4. IEEE, 2047–2052.
- [15] Javier Hernandez, Dan Morris, and Rosalind W Picard. 2018. Using Physiological, Behavioral, and Self-Report Data to Infer Affect in the Workplace. *IEEE Transactions on Affective Computing* (2018).
- [16] John J Hopfield. 1982. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the national academy of sciences* 79, 8 (1982), 2554–2558.
- [17] Soowon Kang, Woohyeok Choi, Cheul Young Park, Narae Cha, Auk Kim, Ahsan Habib Khandoker, Leontios Hadjileontiadis, Heejeung Kim, Yong Jeong, and Uichin Lee. 2023. K-emophone: A mobile and wearable dataset with in-situ emotion, stress, and attention labels. *Scientific data* 10, 1 (2023), 351.
- [18] Diederick P Kingma and Jimmy Ba. 2015. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*.
- [19] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. 2017. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences* 114, 13 (2017), 3521–3526.
- [20] Dani Kiyasseh, Tingting Zhu, and David A Clifton. 2020. Clops: Continual learning of physiological signals. *arXiv preprint arXiv:2004.09578* (2020).
- [21] Evgenia Lazarou and Themis P Exarchos. 2024. Predicting stress levels using physiological data: Real-time stress prediction models utilizing wearable devices. *AIMS neuroscience* 11, 2 (2024), 76.
- [22] Shih-Yu Lee, Caroline Wuertz, Rebecca Rogers, and Yu-Ping Chen. 2013. Stress and sleep disturbances in female college students. *American journal of health behavior* 37, 6 (2013), 851–858.
- [23] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 7871.
- [24] Na Li, Yan Wang, Yijiao Dong, Xiaoxue Chen, Bin Zhang, Xianghua Chen, Kejian Wang, and Ying Sun. 2024. The impact of psychological stress on physiological indicators in healthcare workers: a cross-sectional study. *Frontiers in Public Health* 12 (2024), 1393743.
- [25] Zhizhong Li and Derek Hoiem. 2017. Learning without forgetting. *IEEE transactions on pattern analysis and machine intelligence* 40, 12 (2017), 2935–2947.
- [26] Yunfei Luo, Iman Deznabi, Abhinav Shaw, Natcha Simsiri, Tauhidur Rahman, and Madalina Fiterau. 2024. Dynamic clustering via branched deep learning enhances personalization of stress prediction from mobile sensor data. *Scientific Reports* 14, 1 (2024), 6631.
- [27] Stephen M Mattingly, Julie M Gregg, Pino Audia, Ayse Elvan Bayraktaroglu, Andrew T Campbell, Nitesh V Chawla, Vedant Das Swain, Munmun De Choudhury, Sidney K D'Mello, Anind K Dey, et al. 2019. The tesserae project: Large-scale, longitudinal, in situ, multimodal sensing of information workers. In *2019 CHI Conference on Human Factors in Computing Systems*.
- [28] Amir Muaremi, Bert Arnrich, and Gerhard Tröster. 2013. Towards measuring stress with smartphones and wearable devices during workday and sleep. *Bio-NanoScience* 3, 2 (2013), 172–183.
- [29] German I Parisi, Ronald Kemker, Jose L Part, Christopher Kanan, and Stefan Wermter. 2019. Continual lifelong learning with neural networks: A review. *Neural networks* 113 (2019), 54–71.
- [30] Dushyant Rao, Francesco Visin, Andrei Rusu, Razvan Pascanu, Yee Whye Teh, and Raia Hadsell. 2019. Continual unsupervised representation learning. *Advances in neural information processing systems* 32 (2019).
- [31] Philip Schmidt, Attila Reiss, Robert Duerichen, Claus Marberger, and Kristof Van Laerhoven. 2018. Introducing wesad, a multimodal dataset for wearable stress and affect detection. In *Proceedings of the 20th ACM international conference on multimodal interaction*. 400–408.
- [32] Suzanne C Segerstrom and Gregory E Miller. 2004. Psychological stress and the human immune system: a meta-analytic study of 30 years of inquiry. *Psychological bulletin* 130, 4 (2004), 601.
- [33] Abhinav Shaw, Natcha Simsiri, Iman Deznabi, Madalina Fiterau, and Tauhidur Rahman. 2019. Personalized student stress prediction with deep multitask network. *arXiv preprint arXiv:1906.11356* (2019).
- [34] Herman M Van Praag. 2004. Can stress cause depression? *Progress in neuro-psychopharmacology and Biological Psychiatry* 28, 5 (2004), 891–907.
- [35] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).
- [36] Rui Wang, Fanglin Chen, Zhenyu Chen, Tianxing Li, Gabriella Harari, Stefanie Tignor, Xia Zhou, Dror Ben-Zeev, and Andrew T Campbell. 2014. StudentLife: assessing mental health, academic performance and behavioral trends of college students using smartphones. In *Proceedings of the 2014 ACM international joint conference on pervasive and ubiquitous computing*. 3–14.
- [37] Zifeng Wang, Zizhao Zhang, Sayna Ebrahimi, Ruoxi Sun, Han Zhang, Chen-Yu Lee, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, et al. 2022. Dualprompt: Complementary prompting for rehearsal-free continual learning. In *European conference on computer vision*. Springer, 631–648.
- [38] Zifeng Wang, Zizhao Zhang, Chen-Yu Lee, Han Zhang, Ruoxi Sun, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, and Tomas Pfister. 2022. Learning

to prompt for continual learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 139–149.

[39] Tianyi Zhang, Abdallah El Ali, Alan Hanjalic, and Pablo Cesar. 2022. Few-shot learning for fine-grained emotion recognition using physiological signals. *IEEE Transactions on Multimedia* 25 (2022), 3773–3787.

Towards Continual Visual Anomaly Detection in the Medical Domain

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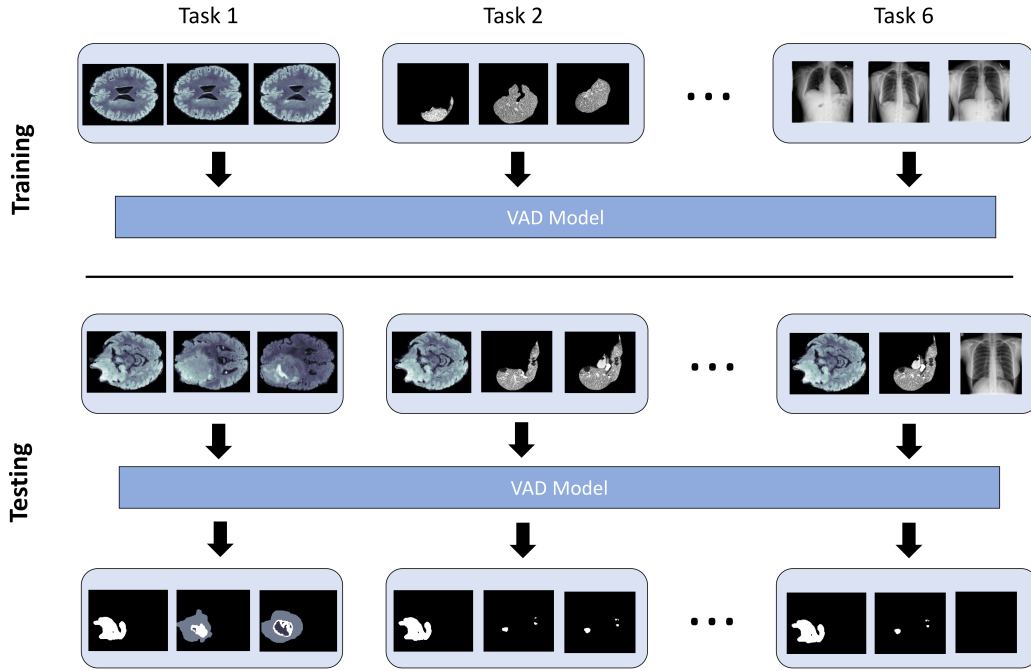


Figure 1: Continual Learning Visual Anomaly Detection for the BMAD medical imaging dataset.

Abstract

Visual Anomaly Detection (VAD) seeks to identify abnormal images and precisely localize the corresponding anomalous regions, relying solely on normal data during training. This approach has proven essential in domains such as manufacturing and, more recently, in the medical field, where accurate and explainable detection is critical. Despite its importance, the impact of evolving input data distributions over time has received limited attention, even though such changes can significantly degrade model performance. In particular,

given the dynamic and evolving nature of medical imaging data, Continual Learning (CL) provides a natural and effective framework to incrementally adapt models while preserving previously acquired knowledge. This study explores for the first time the application of VAD models in a CL scenario for the medical field. In this work, we utilize a CL version of the well-established PatchCore model, called PatchCoreCL, and evaluate its performance using BMAD, a real-world medical imaging dataset with both image-level and pixel-level annotations. Our results demonstrate that PatchCoreCL is an effective solution, achieving performance comparable to the task-specific models, with a forgetting value less than a 1%, highlighting the feasibility and potential of CL for adaptive VAD in medical imaging.



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CCS Concepts

• **Applied computing** → *Life and medical sciences*; • **Computing methodologies** → **Machine learning**.

Keywords

Visual Anomaly Detection, Continual Learning, Medical Imaging

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1 Introduction

Visual Anomaly Detection (VAD) is a critical and important task in Computer Vision and Machine Learning that aims at identifying anomalous patterns in the input images that deviate from the normal distributions. In VAD, the data collection is very critical; indeed, normal data is abundant, and anomalies are very rare and difficult to collect. Because of this, VAD models are trained in a totally unsupervised way by considering only normal data, while in inference, the models are tested on normal and anomalous data. In order to provide explainability of the predictions, the models must return an image-level score but also a segmentation map that pinpoints the anomalous parts of the image. VAD models are employed in a variety of fields spanning from manufacturing [6] to surveillance [19] and the medical domain [1].

In the medical domain, VAD is very delicate, and anomalous patterns may indicate important pathological conditions; this underlines the importance of developing accurate and fast-adapting models in this field. VAD methods have shown strong performance on static scenarios, where a separate model is trained for each considered domain or category. However, VAD models, like every other machine learning model, are prone to Catastrophic Forgetting: when learning a new task, where a task can represent a new category or a new data domain, the model forgets the previous ones.

Continual learning (CL) is a branch of Machine Learning that enables models to incrementally learn and adapt to new tasks without forgetting the previous ones and without needing a complete re-training of the model on new and old data simultaneously (which may be impossible or too demanding).

In this work, we study for the first time the Continual Visual Anomaly Detection task using the BMAD (Benchmarks for Medical Anomaly Detection) [1] dataset, a diverse collection of medical imaging datasets spanning multiple medical domains as anatomical regions and image acquisition modalities. In this work, we adopt PatchCore [18] as Visual Anomaly Detection method, a well-established model for industrial anomaly detection, able to output both image-level anomaly scores and interpretable heatmaps that localize the anomalies in the pixel space. In particular, we test PatchCoreCL, a variant of PatchCore proposed for the continual learning setting [7]. In summary, the main contributions of this paper are:

- We explore for the first time Continual Visual Anomaly Detection setting for the medical field, where new domains are incrementally presented to the VAD model.
- Our results are tested on a CL scenario created from the well-established BMAD dataset and combined with PatchCoreCL (a CL variant of the well-known PatchCore model), obtaining a robust baseline for future research.
- Our results demonstrate that PatchCoreCL is an effective solution, achieving performance comparable to the task-specific models, demonstrating the feasibility for dynamic medical imaging.

Moreover, to promote further research in this new scenario, we provide the code used for the experiments, available on the MoViAD library [3]. The paper is structured as follows: Section 2 provides an overview of the related works in the fields of VAD and CL, Section 3 describes the details about PatchCore and its continual adaptation, while 4 outlines the experimental setting. Section 5 introduces the considered VAD and CL metrics, and Section 6 comments on the obtained results. In the end, Section 7 draws the conclusions of this work and provides some interesting future work directions.

2 Related Work

2.1 Visual Anomaly Detection

A lot of research has been conducted in the VAD domain during the last years, and several VAD models have been developed. Most VAD methods fall into reconstruction-based and feature embedding-based methods.

Reconstruction-based methods use generative models to learn normal image reconstruction during training: large reconstruction errors during inference indicate anomalies. Approaches like AutoEncoders, GANs and Diffusion Models are prominent in this area. An example of these models are [23] and [24].

Feature-based methods rely on data representations generated by pre-trained neural networks. For instance, PatchCore is a feature-based method, which saves features of normal images in a memory bank during training to compare them with test images features during inference.

These techniques balance anomaly detection performance and interpretability by providing both image-level and pixel-level scores and visual heatmaps that pinpoint the image areas depicted as anomalies.

In the last years, VAD research has moved from implementing better models in terms of accuracy towards new challenges such as tackling the computational and memory constraints posed by edge devices [2] [4] as well as dynamic environments, which can benefit from continual learning techniques [7].

2.2 Continual Learning

Traditional Machine Learning models are trained on fixed datasets, but real-world data often differs from the training data distribution. Continual Learning (CL) enables models to adapt to new data without forgetting previous knowledge. Effective CL methods should minimize forgetting, have low memory usage, and be computationally efficient [7].

CL techniques are generally categorized into three main approaches. **Rehearsal-based** approaches, such as Experience Replay [17],

store and reuse past data samples during training. Even if very effective, they are not always applicable, especially in situations when it is not possible to store old data due to data privacy issues or memory limitations.

Regularization-based approaches introduce constraints to the network weights or loss function penalties to retain knowledge of old tasks, often using parameter importance or distillation techniques [10] [12].

Architecture-based approaches modify the model structure to preserve prior knowledge [8] [14].

Experience Replay and, in general, Rehearsal-based approaches are widely considered the most effective strategies to mitigate Catastrophic Forgetting [15], particularly for image classification. However, the best approach may vary by task. For instance, while rehearsal-based methods excel in image classification problem, distillation-based approaches like LwF may be more suitable for the Object Detection [9] problem.

2.3 Continual Visual Anomaly Detection

In recent years, some interest has been shown in the development of Continual Learning Visual Anomaly Detection (CLAD) models. One of the first studies proposed a method called Distribution of Normal Embeddings (DNE) [11], based on the feature distribution of normal training samples from past tasks. However, this method operates only at the image level, limiting its relevance in VAD, where pixel-level information, provided by all the current state-of-the-art approaches, is extremely relevant for interpretability and improving the decision-making process of humans.

Afterwards, [16] evaluates classic VAD models able to provide both image-level and pixel-level information. This work adapts classic VAD methods by using the replay technique (and other more efficient compressed variants) to make them work in the CLAD scenario. Subsequently, [7] expands the previous work by testing several state-of-the-art VAD approaches with the replay technique for many models, such as STFPM [20], EfficientAD [5], and FastFlow [21], showing different forgetting levels based on the chosen architecture.

In particular, for PatchCore, the replay technique is not applicable since their weights are frozen. Therefore, the authors introduced an adaptation to the model to enable it to evolve over time while retaining previous knowledge, while at the same time ensuring that the memory size remains fixed, preventing unbounded memory growth. While other recent works were conducted in the field, like [13], all these studies were conducted by considering industrial datasets such as the MVTec dataset [6].

Though this is a standard practice for the VAD field, focusing only on this specific domain precludes the generalization towards other fields such as the medical field, which could benefit greatly from an adaptive model that can handle new data without the need of retraining. For this reason, in this work we study for the first time the Continual Visual Anomaly Detection problem in the medical domain by considering the BMAD [1] dataset.

3 Methodology

3.1 Preliminaries

PatchCore [18] is a proven and well known method for Visual Anomaly Detection. This method works by considering a CNN network (such as the WideResnet50 [22]) as a feature extractor and a memory bank. The feature extractor is used for extracting the patch feature vectors from different extraction layers in order to get fine and coarse feature details from the training and test input images. The input images of training are thus processed in patches, and the memory bank is constructed by storing the most relevant patches. In particular, during training, the patch feature vectors from the input training images are extracted, and only a subset of them is stored inside the memory bank by considering a KCenter clustering algorithm (called **coreset subsampling** in the original paper). During inference, the patch feature vectors of the input images are compared with a minimax algorithm with the vectors stored in the memory bank. From this patch level anomaly score, it is possible to derive an image-level and a pixel-level anomaly score for the tested images.

3.2 Continual Medical Anomaly Detection Scenario

In the Continual Learning scenario, the model is trained in a sequential way by considering a stream of tasks, where each task is associated with a different dataset category (as illustrated in Figure 1). During inference, the model is tested on every category already encountered during the training up to that point.

This work considers the BMAD dataset [1], a collection of six medical datasets from five different domains for medical anomaly detection. BMAD spans multiple anatomical regions, with six different anomaly detection categories: "Brain_AD", "Liver_AD", "Retina_RESC_AD", "Chest_AD", "Histopathology_AD" and "Retina_OCT2017_AD". BMAD is not inherently built as a continual learning dataset, and, in order to test the continual learning strategies, the dataset is split into a sequence of tasks based on the different medical categories, simulating a continual learning scenario. Each task includes a binary classification between healthy and anomalous samples, both image-level and pixel-level.

Since the task at hand is Visual Anomaly Detection in an unsupervised setting, the training dataset contains only normal samples, while, at test time, both normal (unseen during training) and abnormal samples are evaluated. The stream of tasks is composed of the following categories: "Brain_AD", "Liver_AD", "Retina_RESC_AD", "Chest_AD", "Histopathology_AD" and "Retina_OCT2017_AD".

3.3 Continual PatchCore (PatchCoreCL)

In this work, we consider PatchCore as model for our study. However, given its static nature, classic CL methods like replay cannot work. Indeed, the PatchCore model keeps its weights (the feature extractor) frozen, and only the memory bank containing patches is updated just once using the training data.

Therefore, it is necessary to develop strategies that enable the PatchCore model to work in incremental scenarios. In particular, our study implements the idea proposed in [7] which we refer with the name of *PatchCoreCL* and allows PatchCore to work in the

continual learning setting.

To adapt the PatchCore model to the dynamic continual learning scenario, several modifications were made to its internal memory bank and training procedure. The core idea is to maintain a separate memory bank for each task encountered during training while keeping the total number of stored patch feature vectors fixed to avoid problems of exploding memory. This means that over time, the number of associated vectors for a task is given by the maximum number of samples that can be stored in the memory bank divided by the number of tasks encountered. In practice, this translates in a ever-decreasing number of samples associated to each task. However, instead of applying random sampling on the patches, PatchCoreCL applies a memory bank update procedure that applies separately the coreset reduction step on the stored samples for each task. The training procedure is illustrated in Algorithm 1 while the whole procedure is shown in Figure 2.

Algorithm 1 PatchCore memory bank update for CL method

```

1: Initialize:  $\mathcal{M} \leftarrow$  empty list
2:  $memory\_size \leftarrow 30000|10000$ 
3: for  $i = 0$  to  $N - 1$  do  $\triangleright N = \text{number of tasks}$ 
4:    $p \leftarrow$  patches extracted for task  $i$ 
5:   for  $j = 0$  to  $length(\mathcal{M})$  do
6:      $\mathcal{M}[j] \leftarrow coreset\_subsampling(\mathcal{M}[j], \frac{memory\_size}{i+1})$ 
7:    $m \leftarrow coreset\_subsampling(p, \frac{memory\_size}{i+1})$ 
8:    $\mathcal{M}.append(m)$ 

```

Within the realm of Continual Learning, this methodology is based on a rehearsal strategy for adapting the model to the new tasks, since a portion of old data (or more precisely, portions of the feature maps associated with the images) is stored in the memory banks. During inference, when a sample arrives, the patch feature vectors are compared with every memory bank in the list in order to get an image-level score. The memory bank that ensures the lowest score is then used for calculating the pixel-level score and the final anomaly mask. This is reasonable because the memory banks associated with categories different from the test sample category will cause a higher image-level anomaly score. Thanks to this inference strategy, this Continual Learning adaptation is also able to perform task identification during inference: a critical phase for every continual learning technique.

4 Experimental Setting

4.1 Dataset

As described in Section 3, this work considers the BMAD dataset in a Continual Learning scenario, where the stream of tasks is: "Brain_AD", "Liver_AD", "Retina_RESC_AD", "Chest_AD", "Histopathology_AD" and "Retina_OCT2017_AD". The first three categories are labeled at both image and pixel levels, while the last three are labeled only at the image level.

The images are considered at a resolution of 224 x 224 pixels. Furthermore, due to the large number of samples present for every BMAD category, we decided to consider a maximum of 2000 samples per category for training.

4.2 Compared Strategies

We compare PatchCore under various scenarios to assess its performance in continual learning and compare it with relevant baselines.

Multi-Model: Independently trains one model per task. This is the standard approach in the literature and serves as an upper bound reference, but does not support any form of knowledge transfer or continual adaptation.

Joint-Train: A single model is trained jointly on all tasks. While not a continual method, it offers a performance upper bound assuming full access to all data at once.

Fine-Tuning: A baseline continual learning setup where the model is updated sequentially with no mitigation of forgetting effects. It illustrates the degradation from naive updates.

PatchCoreCL: The proposed PatchCore adaptation to the CL scenario, described in Section 3. The tested versions are:

- PatchCoreCL-30K: considers a memory bank of 30000 patch feature vectors.
- PatchCoreCL-10K: considers a memory bank of 10000 patch feature vectors.

5 Metrics

We evaluate performance across both anomaly detection and continual learning dimensions using standard metrics. Results are reported for image-level and pixel-level anomaly detection, as well as for continual learning-specific metrics.

5.1 Anomaly Detection Metrics

We first assess the anomaly detection capabilities of each strategy using standard Visual Anomaly Detection metrics at both the image and pixel levels.

Image-level metrics assess the model's ability to classify an entire image as anomalous or not. We report the Area Under the Receiver Operating Characteristic Curve (AUROC) and F1 score based on the image-level anomaly score.

Pixel-level metrics evaluate the spatial localization of anomalies via the heatmaps produced by PatchCore. Applicable only for subsets of BMAD that include ground truth segmentation masks, we compute:

- AUROC and F1 score based on per-pixel anomaly scores,
- Precision-Recall (PR) score,
- AU-PRO, the Area Under the Per-Region Overlap curve, which measures overlap between predicted and ground truth anomalous regions.

In the results obtained in the CL scenario, we report all the image-level metrics as the average across all tasks, measured after the model has completed training on the final task. BMAD spans multiple anatomical regions, with six different anomaly detection categories, three of which support pixel-level evaluation ("Brain_AD", "Liver_AD" and "Retina_RESC_AD") while the remaining three are for sample-level assessment only. Therefore, the pixel-level metrics are obtained after training the model on the first three tasks, since only the first three categories in the stream are pixel-level labeled. Furthermore, for the same reason, the relative gap and the average forgetting are calculated at the pixel level.

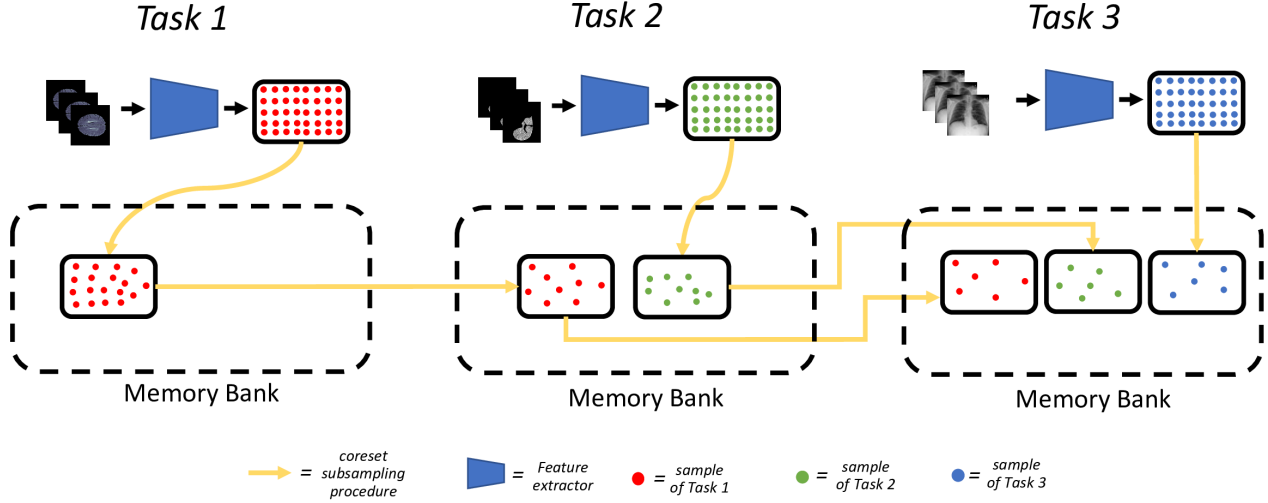


Figure 2: Scheme of PatchCoreCL. When new data arrives, PatchCoreCL applies the procedure of coreset sampling on the new data and creates a memory bank for that task. At the same time, the coreset sampling is also applied to the old memory banks to reduce the number of samples. This allows the number of samples stored in memory doesn't change over time.

5.2 Continual Learning Metrics

To assess continual learning performance, we compute:

Relative Gap (δ): For each continual learning method, we measure the difference between its performance pixel-level F1 ($F1_{CL}$) and the Joint-Train upper bound ($F1_{Joint}$).

Defined as δ :

$$\delta = F1_{Joint} - F1_{CL}$$

Average Forgetting: Mean performance drop across all previously seen tasks when training on a new task. This quantifies the extent to which past knowledge is overwritten in the continual setting. This metric quantifies how much of the previous task knowledge is lost during training.

Defined as F :

$$F = \frac{1}{T-1} \sum_{t=1}^{T-1} \left(\max_{k \in \{1, \dots, t\}} R_{k,t} - R_{T,t} \right)$$

where:

$R_{k,t}$ is the performance (e.g., F1 score) on task t after training on task k ,

$R_{T,t}$ is the performance on task t after training on the final task T .

In this work, we report the Average Forgetting values as percentage values, as done in [7].

When evaluating Continual Learning strategies, it is very important to evaluate the relative gap and the average forgetting at the same time. This comparison is essential, as a model may exhibit low forgetting but substantially lower performance compared to the Joint-Train strategy, which is the upper bound for all the CL approaches.

In addition, for every CL strategy, we reported the Architecture Memory (needed for storing the PatchCore feature extractor) and the Additional Memory (needed for storing the Memory Bank) in terms of MegaBytes (MB). In Continual Learning we are interested

in updating and expanding the model's knowledge over time, avoiding forgetting, while at the same time, we want to obtain this goal with minimal computation and memory overhead. Therefore, these additional metrics are extremely relevant for assessing the practical feasibility and deployment of Continual Learning models in real-world environments.

Table 1: Comparison of PatchCore performance across different learning strategies on the BMAD benchmark. Metrics are reported at both image-level and pixel-level, including AUROC, F1, precision-recall (PR), and AUPRO. We compare Multi-Model, Joint-Train, and the continual learning strategies (Fine-Tuning, PatchCoreCL-10k, PatchCoreCL-30k). The relative gap to the upper bound and average forgetting are also reported to assess performance in continual learning. We also report the Architecture and Additional Memory.

PatchCore		Multi Model	Joint Train	Fine Tuning	PatchCoreCL 10k	PatchCoreCL 30k
Image level	AUROC	0.99	0.82	0.73	0.77	0.81
	F1	0.98	0.86	0.75	0.84	0.85
Pixel level	AUROC	0.94	0.93	0.31	0.93	0.93
	F1	0.64	0.63	0.25	0.62	0.63
	PR	0.70	0.69	0.12	0.67	0.69
	AUPRO	0.78	0.76	0.10	0.75	0.77
Arch. Mem [MB]		275.6	275.6	275.6	275.6	275.6
Add. Mem [MB]		1105.8	184.3	184.3	61.43	184.3
Rel. Gap (δ)		-	-	0.11	0.02	0.01
Avg. Forget. [%]		-	-	52.32	0.80	0.73

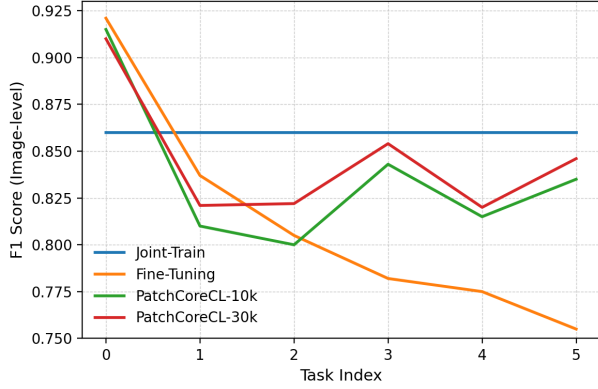


Figure 3: Image-level F1 score for all the tested methods, the task index is chronologically ordered to match the task stream in the continual learning procedure. The F1 is computed for each method as the mean score for all the tasks seen in training up to that point.

6 Results

This section analyzes the results obtained with the different strategies listed in Section 4. In Table 1 all the obtained Anomaly Detection and Continual Learning metrics are reported, along with performance metrics such as additional memory.

As reported in Table 1, the best performing strategy is Multi-Model as expected, where a different model is trained for each BMAD category. This solution is unsurprisingly the best in terms of accuracy, both image- and pixel-level, but implies a large additional memory consumption (1105.8 MB), needed for storing all extracted memory banks (one for each task) while only a single feature extractor is stored.

The Joint-Train strategy, which defines the upper bound for a single-model performance, allows good results at both image- and pixel-level with a reduced memory usage. Instead, the Fine-Tuning approach, as always reported in the literature, is the worst strategy. It achieves the lower results in terms of both image- and pixel-level accuracies.

The two proposed sample-based strategies achieved very promising results, reaching almost the Joint-Train performances on both image- and pixel-level metrics. On image-level metrics, PatchCoreCL-10k suffers a GAP of 0.02 on the F1 score and only 0.01 on the F1 pixel-level, while the PatchCoreCL-30k replicates perfectly the Joint-Train performances.

Very interesting is PatchCoreCL-10k, which uses only one-third of the memory required by Joint-Train, and the obtained performances are not too far from it, with a relative gap of only 0.02 and a very contained forgetting (0.80% against the 0.73% of the PatchCoreCL-30K). The last results show how the chosen memory bank update strategy allows for reducing task interference between tasks and allows for retaining the past knowledge in a very effective way.

In Figure 5.2, the trend of the F1 image-level metric among the considered tasks is reported. As can be visually seen, the Fine-Tuning solution shows the highest forgetting, while the PatchCoreCL-10k and PatchCoreCL-30k solutions show a very small forgetting, a

lower gap to the Joint-Train approach and a good retention of the previous knowledge.

7 Conclusions

In this work, we explored for the first time the Visual Anomaly Detection problem in a Continual Learning scenario for the medical domain, where new domains are incrementally presented to the VAD model.

Our results are tested on a CL scenario created from the well-established BMAD dataset. As the main CL technique, we considered the PatchCoreCL method, providing a robust baseline for future research.

The results show how PatchCoreCL-30K and PatchCoreCL-10K can tackle the problem in a very effective way, showing a very small relative gap with respect to the Joint-Train solution and very low forgetting. Specifically, PatchCoreCL-30K obtains a forgetting value less than 1%. This is accomplished by considering the same amount of memory used in the Joint-Train or even less memory (one-third) in the PatchCoreCL-10K solution. This work paves the way for the development of new Visual Anomaly Detection models under this particular and challenging scenario. Possible future work directions can be focused on improving even more the proposed PatchCore adaptation and on testing even more state-of-the-art Visual Anomaly Detection models on this configuration.

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References

- [1] Jinan Bao, Hanshi Sun, Hanqiu Deng, Yinsheng He, Zhaoxiang Zhang, and Xingyu Li. 2023. BMAD: Benchmarks for Medical Anomaly Detection. *arXiv preprint arXiv:2306.11876* (2023).
- [2] Manuel Barusco, Francesco Borsatti, Davide Dalle Pezze, Francesco Paissan, Elisabetta Farella, and Gian Antonio Susto. 2025. PaSTe: Improving the Efficiency of Visual Anomaly Detection at the Edge. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR) Workshops*. 4026–4035.
- [3] Manuel Barusco, Francesco Borsatti, Arianna Stropeni, Davide Dalle Pezze, and Gian Antonio Susto. 2025. MoViAD: A Modular Library for Visual Anomaly Detection. *arXiv:2507.12049 [cs.CV]* <https://arxiv.org/abs/2507.12049>
- [4] Manuel Barusco, Lorenzo D’Antoni, Davide Dalle Pezze, Francesco Borsatti, and Gian Antonio Susto. 2025. Memory Efficient Continual Learning for Edge-Based Visual Anomaly Detection. *arXiv:2503.02691 [cs.CV]* <https://arxiv.org/abs/2503.02691>
- [5] Kilian Batzner, Lars Heckler, and Rebecca König. 2024. EfficientAD: Accurate Visual Anomaly Detection at Millisecond-Level Latencies. *arXiv:2303.14535 [cs.CV]* <https://arxiv.org/abs/2303.14535>
- [6] Paul Bergmann, Michael Fauser, David Sattlegger, and Carsten Steger. 2019. MVTEC AD – A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 9584–9592.
- [7] Nikola Bugarin, Jovana Bugaric, Manuel Barusco, Davide Dalle Pezze, and Gian Antonio Susto. 2024. Unveiling the anomalies in an ever-changing world: A benchmark for pixel-level anomaly detection in continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4065–4074.
- [8] Chrisantha Fernando, Dylan Banarse, Charles Blundell, Yori Zwols, David Ha, Andrei A. Rusu, Alexander Pritzel, and Daan Wierstra. 2017. PathNet: Evolution

- Channels Gradient Descent in Super Neural Networks. arXiv:1701.08734 [cs.NE] <https://arxiv.org/abs/1701.08734>
- [9] Linting Guan, Yan Wu, Junqiao Zhao, and Chen Ye. 2018. Learn to detect objects incrementally. In *2018 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 403–408.
- [10] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. 2017. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences* 114, 13 (March 2017), 3521–3526. doi:10.1073/pnas.1611835114
- [11] Wujin Li, Jiawei Zhan, Jinbao Wang, Bizhong Xia, Bin-Bin Gao, Jun Liu, Chengjie Wang, and Feng Zheng. 2022. Towards continual adaptation in industrial anomaly detection. In *Proceedings of the 30th ACM International Conference on Multimedia*. 2871–2880.
- [12] Zhizhong Li and Derek Hoiem. 2017. Learning without Forgetting. arXiv:1606.09282 [cs.CV] <https://arxiv.org/abs/1606.09282>
- [13] Jiaqi Liu, Kai Wu, Qiang Nie, Ying Chen, Bin-Bin Gao, Yong Liu, Jinbao Wang, Chengjie Wang, and Feng Zheng. 2024. Unsupervised continual anomaly detection with contrastively-learned prompt. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 38. 3639–3647.
- [14] Arun Mallya and Svetlana Lazebnik. 2018. PackNet: Adding Multiple Tasks to a Single Network by Iterative Pruning. arXiv:1711.05769 [cs.CV] <https://arxiv.org/abs/1711.05769>
- [15] Lorenzo Pellegrini, Gabriele Graffieti, Vincenzo Lomonaco, and Davide Maltoni. 2020. Latent Replay for Real-Time Continual Learning. arXiv:1912.01100 [cs.LG] <https://arxiv.org/abs/1912.01100>
- [16] Davide Dalle Pezze, Eugenia Anello, Chiara Masiero, and Gian Antonio Susto. 2022. Continual learning approaches for anomaly detection. *arXiv preprint arXiv:2212.11192* (2022).
- [17] David Rolnick, Arun Ahuja, Jonathan Schwarz, Timothy Lillicrap, and Gregory Wayne. 2019. Experience Replay for Continual Learning. In *Advances in Neural Information Processing Systems*, H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett (Eds.), Vol. 32. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2019/file/fa7cdfad1a5aaf8370ebeda47a1ff1c3-Paper.pdf
- [18] Karsten Roth, Latha Pemula, Joaquin Zepeda, Bernhard Schölkopf, Thomas Brox, and Peter Gehler. 2022. Towards Total Recall in Industrial Anomaly Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 14318–14328.
- [19] K. K. Santhosh, D. P. Dogra, and P. P. Roy. 2020. Anomaly Detection in Road Traffic Using Visual Surveillance: A Survey. *ACM Comput. Surv.* 53, 6, Article 119 (Dec. 2020), 26 pages. doi:10.1145/3417989
- [20] Guodong Wang, Shumin Han, Errui Ding, and Di Huang. 2021. Student-teacher feature pyramid matching for anomaly detection. *arXiv preprint arXiv:2103.04257* (2021).
- [21] Jiawei Yu, Ye Zheng, Xiang Wang, Wei Li, Yushuang Wu, Rui Zhao, and Liwei Wu. 2021. FastFlow: Unsupervised Anomaly Detection and Localization via 2D Normalizing Flows. arXiv:2111.07677 [cs.CV] <https://arxiv.org/abs/2111.07677>
- [22] Sergey Zagoruyko and Nikos Komodakis. 2017. Wide Residual Networks. arXiv:1605.07146 [cs.CV] <https://arxiv.org/abs/1605.07146>
- [23] Vitjan Zavrtanik, Matej Kristan, and Danijel Skočaj. 2021. DRAEM – A discriminatively trained reconstruction embedding for surface anomaly detection. arXiv:2108.07610 [cs.CV] <https://arxiv.org/abs/2108.07610>
- [24] Vitjan Zavrtanik, Matej Kristan, and Danijel Skočaj. 2021. Reconstruction by inpainting for visual anomaly detection. *Pattern Recognition* 112 (2021), 107706. doi:10.1016/j.patcog.2020.107706

What is Wrong with Continual Learning in Medical Image Segmentation?

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Abstract

Continual learning (CL) is crucial for advancing medical image segmentation. In real-world clinical scenarios, datasets arrive sequentially, are often incomplete, and are subject to strict privacy constraints, making traditional static training approaches impractical. While many CL methods focus on mitigating catastrophic forgetting, they frequently overlook the broader requirements of scalability and compatibility with real-world workflows. Moreover, most existing methods are not robust to distribution shifts and require domain labels. We present *UNEG (U-Net Expert Gate)*, a simple yet highly effective multi-model benchmark for continual learning in medical image segmentation. UNEG avoids catastrophic forgetting by maintaining separate task-specific segmentation models and using an autoencoder-based oracle to dynamically identify the most appropriate model during inference. Unlike complex methods, UNEG adheres to key constraints necessary for deployment in clinical environments: privacy compliance, robustness across domains, scalability, and generalizability. Through a systematic comparison, we show that UNEG satisfies all these constraints, making it an ideal baseline for evaluating the effectiveness of other CL methods. This work not only highlights the limitations of existing methods but also redefines the priorities for future research to ensure clinical applicability and deployment readiness.

CCS Concepts

• **Applied computing** → **Health informatics**; • **Computing methodologies** → **Lifelong machine learning**; **Image segmentation**.

Keywords

continual learning, lifelong learning, image segmentation

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1 Introduction

Supervised Deep Learning (DL) has emerged as the dominant approach for medical image segmentation, delivering state-of-the-art performance in static training setups where datasets are fully accessible and utilized simultaneously. However, this paradigm is increasingly at odds with real-world clinical workflows, where datasets arrive incrementally, are often constrained by privacy regulations, and exhibit domain shifts due to varying acquisition protocols or imaging hardware [44]. The medical imaging community is growing aware of the discrepancies between how DL algorithms are evaluated and the performance drop that occurs in real clinical settings. This manifests in an increased interest in alternative training and evaluation protocols such as federated [7, 31, 55] and continual [6, 14, 20, 30, 35, 37, 42, 43, 49, 57, 58] learning. Continual learning (CL) in particular addresses the possibility that datasets from different domains arrive sequentially and are only accessible during a certain time interval, as illustrated in Figure 1.

Despite significant advancements in CL, existing methods fall short of addressing the unique requirements of medical imaging. Many approaches focus exclusively on preventing *catastrophic forgetting*, where performance on earlier tasks degrades as new tasks are introduced, without sufficiently considering the challenges posed by domain variability, scalability, or privacy compliance. Yet the actual goal of continual learning should be achieving *positive backward transfer*, which occurs when the performance on data from early domains *improves* as training continues. Furthermore, the complexity of current CL methods often limits their clinical applicability, as they require intricate architectural modifications, domain-specific parameters, or extensive retraining, making them impractical for routine use. To address these challenges, it is essential to consider several critical constraints for successful deployment in clinical environments. These constraints include:

- **Privacy Compliance:** Ensuring that data is not stored or revisited, adhering to ethical and legal medical standards.
- **Robustness Across Domains:** Generalizing to new domains without relying on domain metadata or labels.

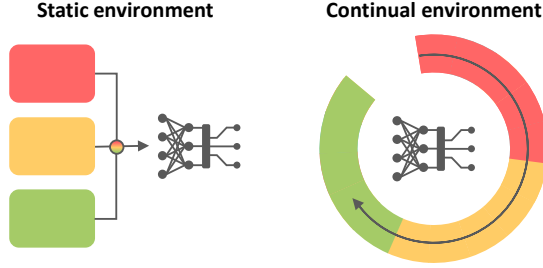


Figure 1: Static vs. continual protocols for training and evaluating machine learning models.

- **Scalability:** Maintaining computational efficiency, ensuring that the complexity of the models does not increase proportionally with the number of tasks.
- **Compatibility:** Integration with state-of-the-art architectures, allowing for robust performance on complex tasks.

We introduce *UNEG (U-Net Expert Gate)*, a practical and robust multi-model framework for CL in medical image segmentation. UNEG adopts a task-specific architecture that mitigates catastrophic forgetting by keeping separate segmentation models for each domain. An autoencoder-based oracle dynamically selects the most suitable model during inference, enabling effective handling of domain shifts without requiring prior knowledge of domain identity.

This simple yet powerful solution addresses all the constraints for clinical deployment. UNEG ensures **privacy compliance** by avoiding the storage or revisiting of previous training data and demonstrates **robustness across domains** by using a reconstruction-based oracle that enables, in contrast to domain-specific methods, effective generalization to unseen domains. **Scalability** is ensured, as model complexity remains manageable, preventing the computational overhead typically encountered in task-specific methods. Finally, UNEG offers **model compatibility**, integrating seamlessly with advanced architectures like nnU-Net, and providing robust performance on complex medical imaging tasks.

Existing CL methods often benchmark against static training setups, which assume all data is available simultaneously – a condition rarely met in real-world applications. Furthermore, many methods rely on domain labels during inference or introduce task-specific architectural modifications. UNEG provides a practical, efficient, and scalable solution that checks all the necessary boxes for deploying continual learning in medical image segmentation while overcoming the limitations of existing methods.

2 Problem Formulation

In a continual learning setting, the database consists of N datasets $\{X_i, Y_i\}_{i \leq N}$. Each dataset comprises samples (x, y) , where x is an input and y is the corresponding annotation. This work focuses on image segmentation, where each sample is a pair consisting of an image x and a segmentation mask y . The goal is to train a model $\mathcal{F}$ that performs well across all datasets, which arrive sequentially and are only available for a limited period. Figure 1 illustrates the

sequential data availability inherent in a continual learning protocol. The model $\mathcal{F}$ is trained iteratively with $\{X_1, Y_1\}$, $\{X_2, Y_2\}$, and so on, until it incorporates information from all available datasets. Unlike static setups, samples $(x, y) \in \{X_1, Y_1\}$ share specific characteristics stemming from their generation processes, which may differ from those in other training stages.

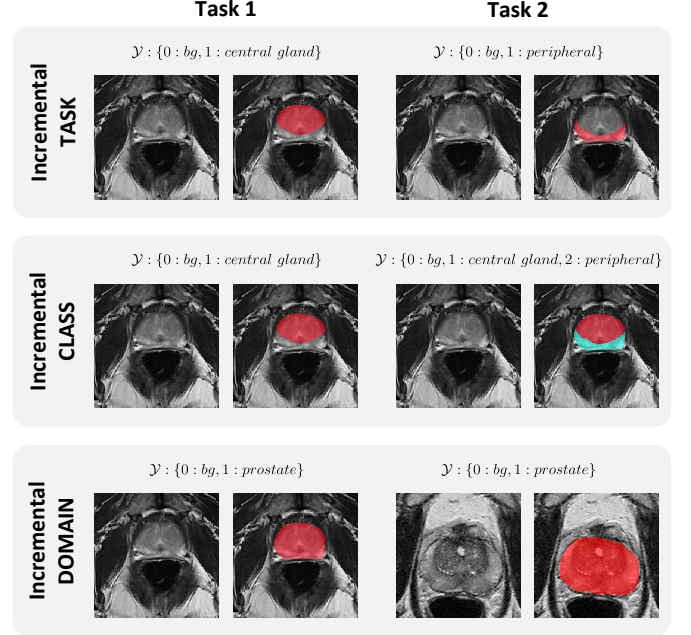


Figure 2: Three CL scenarios, exemplified for the case of prostate segmentation on T2-weighted MRIs.

2.1 Continual Learning Scenarios

Differences in sample distributions can manifest in several ways. van de Ven et al. [56] introduce three continual learning scenarios, as demonstrated in Figure 2 for prostate segmentation.

In the *incremental task scenario*, both the input and label spaces can vary, typically with task identity information provided. For example, a network might segment the prostate’s central gland for task 1 and the peripheral zone for task 2. However, this scenario is rare in image segmentation. The *incremental class scenario* involves incrementally adding new classes to the label space. For instance, dataset 1 enables learning the central gland class, while dataset 2 introduces the peripheral zone class, resulting in a model capable of segmenting both classes. This scenario, while interesting, is only meaningful in specific contexts and requires architectural modifications each time a new class is introduced. Finally, in the *incremental domain scenario*, the semantic knowledge remains constant, but differences in image characteristics arise, such that $X_i \simeq X_j$ and $Y_i = Y_j$. This scenario is the most prevalent in medical imaging, occurring when systems must incrementally learn from images acquired from different sources or at different times. Thus, this study focuses on incremental domain learning.

2.2 Domain Identity and its Implications

A *domain* $\mathcal{D}$ refers to a set of image characteristics unique to the acquisition source, independent of the content of interest. For example, images from one MR scanner may have different contrast levels than those from another scanner, and the region of interest (ROI) may vary slightly based on acquisition protocols.

In the incremental domain scenario, domain identity information can be either available or unavailable during inference. We define two settings based on this distinction. In the simpler *Domain Knowledge* setting, test inputs are provided as (x, i) , where i specifies that $x \in X_i$. For instance, if the model is trained on data from three scanners, metadata on the scanner used for each image is available during inference. In the *No Domain Knowledge* setting, such information is unavailable during testing.

The availability of domain identity labels enables the use of *domain-specific parameters* that are not shared across domains, requiring only the *shared parameters* to be trained continually. For classification, this typically involves sharing the feature extraction layers while maintaining domain-specific layers for predictions.

2.3 Domain Knowledge Assumptions

There are significant limitations to the *Domain Knowledge* scenario. First, the model can only operate on data from domains encountered during training. Second, assuming domain labels are available during deployment is often unrealistic. Metadata encoding acquisition information can vary significantly, even within a single institution, and may be unavailable due to anonymization processes in tele-radiology systems. Furthermore, in dynamic settings with multiple varying factors, identifying the most relevant factor can be nontrivial. This motivates the need for robust continual learning solutions that can generalize to new domains without relying on explicit domain labels during inference.

2.4 The Challenge of Catastrophic Forgetting

The most straightforward approach to training a model continuously is sequential fine-tuning, where the model is updated as new data becomes available. However, if samples from different datasets stem from distinct distributions, this violates the assumption of independent and identically distributed (i.i.d.) data required by stochastic gradient descent (SGD). A common source of variability in diagnostic radiology arises from differences in acquisition protocols or equipment vendors, a phenomenon referred to as *domain shift* or *domain interference* [15].

Neural networks trained with SGD on sequentially arriving datasets often overfit to the domain characteristics of the most recent data. This adaptation leads to a significant drop in performance on earlier datasets, a phenomenon known as *catastrophic forgetting* [36]. On the other hand, overly constraining the model to preserve past knowledge can hinder its ability to adapt to new data. Therefore, special care must be taken during training to ensure that the final model performs well across all datasets encountered throughout the learning process.

While mitigating catastrophic forgetting and maintaining model plasticity are critical, we argue that these are only foundational objectives for continual learning. The broader goal should be to

ensure robust performance across diverse data distributions while enabling effective knowledge integration from new domains.

2.5 Mitigating Catastrophic Forgetting

Several strategies have been developed to address catastrophic forgetting, with their applicability to medical imaging explored in classification [20, 24, 30, 49] and segmentation [35, 37, 42, 43, 45–47, 57, 58]. These methods are typically evaluated against the baseline of sequential fine-tuning and other continual learning approaches, as well as the upper bound of static training, which assumes the availability of all training data. However, a common oversight is the lack of comparison against a simple multi-model benchmark, which involves maintaining one model per domain and could potentially outperform the proposed methods.

In scenarios where *Domain Knowledge* is available, allowing domain-specific information to be used during testing, a straightforward way to mitigate catastrophic forgetting is to maintain separate models $\mathcal{F}_i$ for each domain $\{X_i\}_{i \leq N}$. In this setup, model $\mathcal{F}_1$ is trained solely on data $\{X_1, Y_1\}$, model $\mathcal{F}_2$ is initialized with the parameters of $\mathcal{F}_1$ and fine-tuned on $\{X_2, Y_2\}$, and so on. Although only the final model $\mathcal{F}_N$ integrates information from all domains, this strategy avoids catastrophic forgetting entirely. Each model $\mathcal{F}_i$ retains its performance on the domain $\{X_i, Y_i\}$, without sacrificing plasticity, as there are no constraints on parameter updates.

Despite its simplicity, this multi-model approach has been scarcely explored in medical imaging. Only three studies [22, 42, 58] explicitly compare against this benchmark. In these cases, either the multi-model approach is considered an upper bound, or the improvement over it is marginal. Moreover, some studies evaluate continual learning methods using metrics such as *backward transfer* or *forgetting*, which compare the final performance on each domain after training on all domains to the performance immediately following training on the respective domain [14, 41, 43, 49]. However, reports of *positive backward transfer* or *negative forgetting*, where performance improves after learning additional domains, are rare.

This raises concerns about the practicality of these methods in clinical workflows. If a model does not improve after continual training, the primary advantage of maintaining a single model over multiple models is reduced storage requirements, which is rarely a limiting factor in clinical environments where robustness and reliability are paramount. Additionally, many continual learning methods implicitly assume the availability of domain identity information during testing. In scenarios without *Domain Knowledge*, or when the model must generalize to previously unseen domains, **many existing approaches are not directly applicable.**

2.6 Proposed Evaluation Framework

We argue that all proposed CL methods should be evaluated against a *multi-model benchmark*, as it represents a straightforward and robust solution to the catastrophic forgetting problem, setting a natural lower bound for performance. Unlike many continual learning methods that necessitate specialized training schemes or architectural modifications, this benchmark maintains the simplicity of the original training procedure and model architecture.

Figure 3 outlines the proposed evaluation framework. - **If domain identity information is available during inference, a**

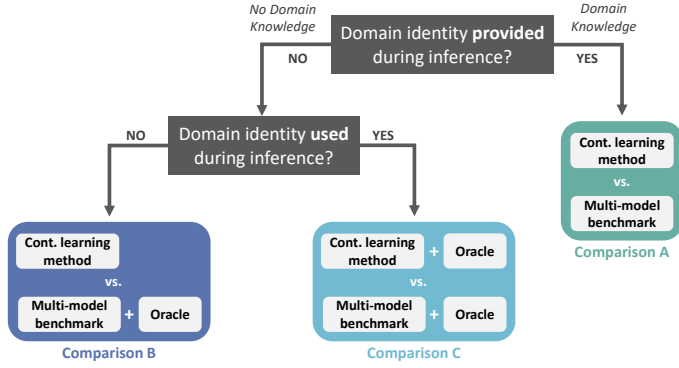


Figure 3: Proposed evaluation framework for continual learning methods. - Comparison A: When domain identity information is available, methods are evaluated against a multi-model benchmark maintaining one model per domain. - Comparison B: When domain identity is unavailable, the benchmark leverages an oracle to identify the closest domain, while the method is evaluated without domain knowledge. - Comparison C: If the method requires domain information, it can also use the oracle for evaluation. If the benchmark outperforms the proposed method, the latter is only preferable under constraints such as memory limitations.

direct comparison with the benchmark can be performed. - **If the proposed method relies on domain information**, the evaluation should consider both the continual learning method and the benchmark using domain identity inferred by an *oracle* for incoming data. - **If the method does not require domain information**, the oracle is used solely by the benchmark, enabling a fair comparison in scenarios where domain knowledge is unavailable.

If the proposed method outperforms the benchmark, it can be considered an improvement. However, if it performs worse, its adoption is justifiable only under specific constraints, such as limited memory or other application-specific requirements.

3 Related Work

This section provides an overview of common continual learning strategies, followed by a discussion of recent research in semantic segmentation and its applications in diagnostic radiology.

3.1 Catastrophic Forgetting Prevention

Continual learning strategies can be broadly categorized as follows:

Rehearsal methods store a subset of previous task examples and interleave them periodically during training [3, 46, 48, 51]. While effective in practice, they scale poorly with increasing domain numbers and require the ability to store training data.

Pseudo-rehearsal methods simulate rehearsal effects without explicitly storing data. These approaches leverage generative models [11, 53], distillation losses to match prior outputs [29, 32], or hybrid memory components [4, 23]. Care must be taken to ensure generative models do not regenerate original data [62].

Sparse-connectivity methods minimize representational overlap between tasks, as overlap is a key cause of forgetting [19, 26].

Techniques such as task-specific masks [16, 34] stabilize performance by isolating network regions but require task identity at inference time.

Network growing approaches add trainable parameters to accommodate new tasks. Methods range from layer or neuron additions [18, 59] to merging separate models [39, 52]. While capacity limitations are mitigated, these methods result in linear space growth and demand continuous architecture adjustments.

Regularization approaches identify important parameters post-training and penalize deviations during future tasks [1, 25, 47]. Variations arise primarily in importance calculation techniques.

Bayesian approaches mitigate forgetting through Bayesian Neural Networks [12, 40], though at significant computational cost.

Other notable strategies include learning **domain-invariant features** [13] and maintaining **task-specific batch normalization parameters** [50].

3.2 CL for Semantic Segmentation

Unlike classification tasks, semantic segmentation faces unique challenges: output masks must preserve geometric properties and connected components to resemble the ground truth. Sequential training can degrade these characteristics, leading to a more significant performance gap between static and continual learning results. While most continual learning methods are designed for classification, adaptations have emerged for semantic segmentation. **Pseudo-rehearsal strategies** include saving image statistics for pseudo-example generation [61]. Shmelkov et al. [54] proposed a distillation loss to mitigate forgetting in object detection. Building on this, Michieli et al. [38] incorporated feature-level distillation but found limited benefits over output-level distillation. Cermelli et al. [10] improved segmentation by accounting for background pixel changes across domains. **Regularization-based methods** include the use of saliency maps for parameter importance [41], while **sparse-connectivity approaches** employ task-specific masks [35].

3.3 Continual Learning in Medical Imaging

Continual learning in medical imaging predominantly adapts regularization and pseudo-rehearsal strategies, with significant focus on brain MRI segmentation.

Regularization methods have been applied to glioma [57] and white matter lesion segmentation [6], as well as chest X-ray classification [30]. Özgün et al. [43] adapted Memory-Aware Synapses (MAS) for brain segmentation, improving importance calculation.

Rehearsal strategies have been evaluated for humerus and scapula segmentation in MRIs [42] and chest CT classification [20]. Venkataramani et al. [58] explored continual lung segmentation using a memory-based approach.

Domain-specific methods address scanner-induced variations in brain MR images. Karani et al. [22] utilized shared convolutional layers with domain-specific batch normalization, outperforming multi-network approaches but requiring access to all domains.

Domain-invariant feature learning aims to improve generalizability. Kim et al. [24] maximized mutual information between features and outputs, achieving superior results for tuberculosis classification on chest X-rays. Elskhawey et al. [14] disentangled

domain-dependent features using adversarial training, while Ravishankar et al. [49] proposed feature transformer networks for domain adaptation.

Medical image segmentation is particularly challenging due to the geometric heterogeneity of ground truth masks across datasets. Most studies focus on a single region of interest, whereas we report results for prostate, hippocampus, and right ventricle.

4 Methods

This section introduces our approach to model selection in continual learning settings, highlighting the role of domain knowledge, the use of multi-model benchmarks, and a proposed oracle for domain identification. The flowchart in Figure 3 summarizes the different evaluation scenarios, and Figure 4 illustrates how model states can be selected during inference.

4.1 Model State Selection in Continual Learning

If model states are stored after each training stage, there are three primary methods for selecting parameters, depending on the availability of domain knowledge at inference time.

Domain Knowledge Setting: When domain identity labels are available, the model state corresponding to the specific domain can be directly utilized during inference (Figure 4, top).

No Domain Knowledge Setting with Oracle: In scenarios where domain identity is not provided, an oracle determines the most suitable model state based on the properties of the test image (Figure 4, middle).

Single-Model Setting: Alternatively, a single model state trained sequentially on all domains can be employed for all predictions, avoiding domain-specific state selection (Figure 4, bottom).

CL methods are evaluated against a *multi-model benchmark* which maintains a separate model for each domain. This setup eliminates catastrophic forgetting and loss of plasticity, ensuring optimal performance by directly associating test samples with their respective domains.

In the *Domain Knowledge* scenario, the appropriate model is selected using domain labels. Without domain knowledge, the benchmark provides an upper bound for performance, where an oracle selects the most suitable model state. However, this benchmark has two significant drawbacks: (1) Space complexity grows linearly with the number of domains, and (2) each domain-specific model does not leverage information from subsequent tasks, preventing backward transfer.

For simplicity, we assume that all parameters are domain-dependent in this work. However, selective sharing of parameters (e.g., shared encoders with domain-specific heads) can be explored, as commonly done in continual learning methods.

4.2 Autoencoder-Based Oracle

To address the *No Domain Knowledge* setting, we propose an oracle-based approach inspired by Aljundi et al. [2]. The oracle utilizes autoencoder networks to identify the domain of a given test image during inference.

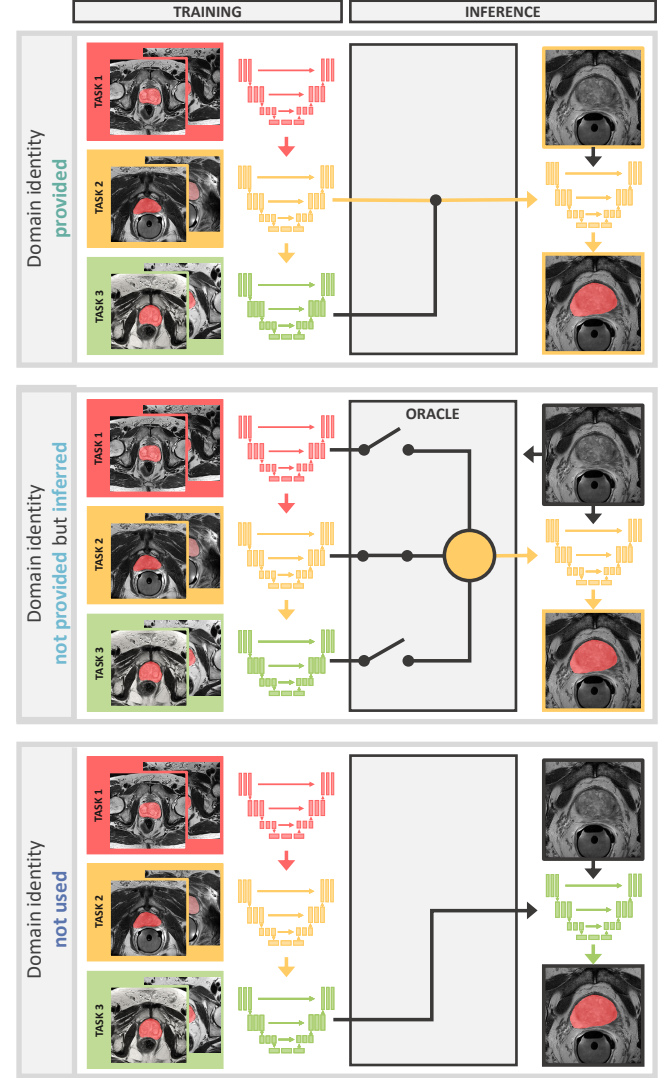


Figure 4: Overview of model selection strategies in continual learning. Top: Domain information is provided at inference time. Middle: A domain detection oracle selects the appropriate model state. Bottom: A single model is used for all predictions. Shared parameters are updated sequentially, while domain-dependent states are saved at each training stage.

An autoencoder $\mathcal{A}$ is a neural network designed to reconstruct its input by learning the mapping $\mathcal{A} : X \rightarrow X$. It minimizes the mean-squared error (MSE) between the input x and its reconstruction $\mathcal{A}(x)$ for n samples, as defined in Equation 1:

$$\mathcal{L}_{MSE} = \frac{1}{n} \sum_{i=1}^n (x_i - \mathcal{A}(x_i))^2. \quad (1)$$

For each domain encountered during training, an autoencoder $\mathcal{A}_i$ is trained to reconstruct images from that specific domain. Training can occur in parallel with segmentation model training, depending on hardware constraints. During inference, a test image x is reconstructed using all previously trained autoencoders $\{\mathcal{A}_i\}_{i \leq N}$. The domain corresponding to the autoencoder with the smallest reconstruction error is selected, and the corresponding segmentation model $\mathcal{F}_i$ is used to generate the prediction $\hat{y}$:

$$\hat{y} = \mathcal{F}_i(x), \text{ where } i = \operatorname{argmin}_i \mathcal{L}_{MSE}(x, \mathcal{A}_i(x)). \quad (2)$$

Autoencoders typically consist of an *encoder*, which reduces the spatial dimensions of the input, and a *decoder*, which reconstructs the input from the reduced representation. Unlike Aljundi et al. [2], who reconstructed *AlexNet* features, we propose leveraging the same architecture used for segmentation tasks (e.g., U-Net). U-Net structures are well-suited for medical image segmentation due to their encoder-decoder design, requiring no additional architectural modifications. To adapt U-Net for domain identification, we remove the final layer responsible for discretizing outputs into class labels. Instead, we train the network to minimize the MSE loss, enabling the reconstruction of input images.

5 Experimental Setup

In the following, we describe our data corpus for three different image segmentation problems. We also state details on the architecture and training procedure for all models.

5.1 Data

We evaluate the proposed approach across three anatomies in MRIs, namely the prostate, hippocampus, and right ventricle. For prostate, we use T2-weighted MRIs from five different sites [33]. The datasets are different in terms of manufacturer and acquisition settings, and each contains 12 to 30 cases. We train in the order $BIDMC \rightarrow I2CVB \rightarrow HK \rightarrow UCL \rightarrow RUNMC$. The delineations encompass both the central gland and the peripheral zone. The hippocampus corpus consists of the *Multi-contrast submillimetric 3 Tesla hippocampal subfield segmentation* (henceforth referred to as *Dryad*) dataset [28], the *Harmonized Hippocampal Protocol* dataset [60] (*HarP* for short) and the data released as part of the *Medical Segmentation Decathlon (DecathHip)* [5]. We train in the order $DecathHip \rightarrow Dryad \rightarrow HarP$. The segmentation masks cover the posterior and anterior hippocampus. For right ventricle segmentation, we use the data released for the *Multi-Centre, Multi-Vendor and Multi-Disease (M&M) Cardiac Segmentation Challenge* [8], which contains two datasets with 75 samples each, the first acquired with *Siemens* scanners, the second with *Philips*.

5.2 Segmentation nnU-Net

We use the patch-based, full-resolution variation of the *nnU-Net* [21]. One model is trained per anatomy, and we perform 250 epochs per dataset. The architecture and training configuration, such as the patch size, are automatically configured by the framework. As we perform continual training, the settings selected for the first dataset are maintained for subsequent data of the same anatomy. The patch sizes used are [28, 256, 256] for prostate, [40, 56, 40] for hippocampus, and [14, 256, 224] for the right ventricle.

5.3 Continual Learning Baselines

We compare our proposed benchmark to three popular continual learning methods. We use the implementations and default hyperparameters of the *Lifelong nnU-Net* framework [17]. All methods are trained in 3D full-resolution with the same configurations as stated in the previous section. We explore *EWC* (Elastic Weight Consolidation)[25] with a $\lambda_{EWC} = 0.4$, *LwF* (Learning without Forgetting)[32] with a temperature of 2 and *MiB* (Modeling the Background)[10], with the alpha parameter of 0.9 and KD loss weighting to 1. These are the parameters suggested by default. We additionally compare to sequential learning without using any mechanism for knowledge preservation and to the upper bound of static, *joint* training where we use all training data at once.

5.4 Autoencoder Architectures

The *Expert Gate* method by Aljundi et al. [2] proposes extracting features from a pre-trained *AlexNet model* [27] and reconstructing them with a two-layer CNN. We test this approach, though we do not believe it is the most suitable for medical images, and refer to this method as **AlexNet z-CNN**. Since AlexNet is trained for RGB image classification, we use two-dimensional slices. Replicating the channel 3 times and feeding it through the AlexNet results in a feature volume of 256 channels with reduced spatial resolution.

As autoencoders follow a similar architecture to U-Nets and other popular segmentation models, we propose mimicking the same architecture as used for segmentation. In the case of the nnU-Net, each model is already configured for the specific particularities of the data. We simply modify the last layer to not discretize the logits with a softmax function; and minimize the reconstruction error to the input instead of the segmentation loss. This is our proposed **UNEG (U-Net Expert Gate)** oracle. We also try an alternative autoencoder that directly reconstructs the input images, namely that offered by the **MONAI** framework [9] which consists of convolution, instance normalization and PReLU blocks. Different to the nnU-Net autoencoder, there is no change in spatial resolution.

To assess whether it is the features or the model which are the most relevant, we experiment as well with using *features from the corresponding nnU-Nets*, reconstructed with a 2-layer CNN autoencoder. We call this method, which also works in three dimensions, **nnU-Net z-CNN**. Similarly, we try out a **CNN** network to reconstruct the images directly. In both cases, the features are taken from the last decoder block.

All autoencoders are trained to minimize the MSE between the reconstruction and the input for 250 epochs. During inference/evaluation, we use the segmentation network that corresponds to the autoencoder with the smallest reconstruction error.

6 Results

We first compare the proposed **UNEG** benchmark to several CL methods. We then perform an ablation study where we explore alternatives for the autoencoder oracle. Finally, we look at a few visual examples of reconstructions from the nnU-Net autoencoders.

6.1 Comparison to CL methods

Figure 6 visualizes the results for prostate segmentation. The first boxplot shows the upper bound of a model trained statically with

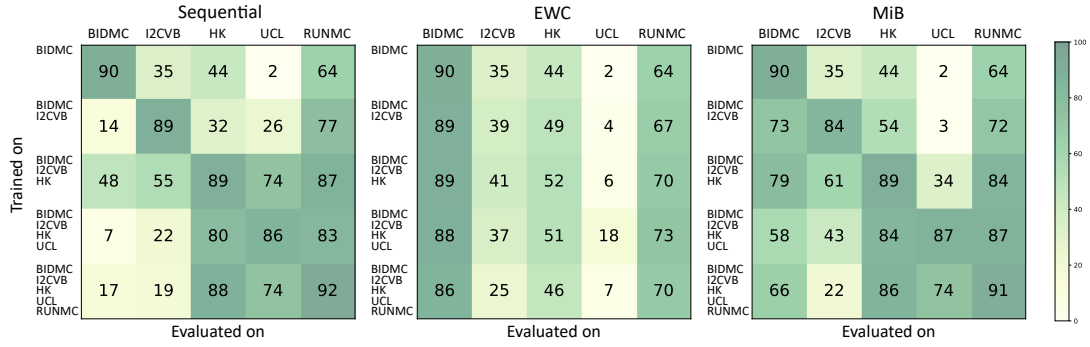


Figure 5: Mean Dice of five prostate segmentation tasks, after training with each of five stages. The diagonal from the upper left to lower right corners shows the score after training with the corresponding training data.

all data. For prostate segmentation, a Dice of around 90% is expected in a static scenario. We then see the results for training a model sequentially without forgetting prevention, where the scores are distributed across the performance spectrum. Three continual learning methods follow, namely EWC, LwF and MiB, the latter of which performs best. The multi-model *UNEG* benchmark performs considerably better than continual learning approaches, though there is still a gap in performance to static training.

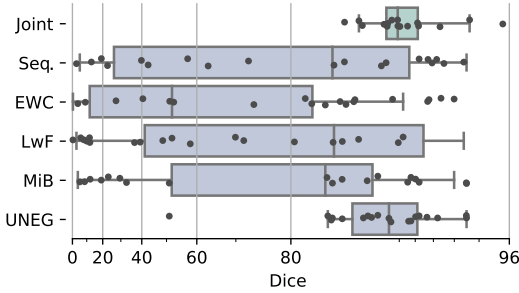


Figure 6: Dice scores of the final model state on test data from five prostate segmentation datasets.

Figure 5 provides additional insight into how the methods perform at different stages. For sequential training, we notice the performance deterioration typical of catastrophic forgetting. *MiB* somewhat reduces this effect, but the final model produces low-quality segmentations for the initial stages. *EWC* instead displays a different behavior: performance remains high for the first task, but the model is constrained in its ability to capture new knowledge. A similar but more pronounced behavior occurs for hippocampus segmentation. Figure 7 visualizes these results. For all single-model methods, we see a clear separation between samples that are correctly segmented and those for which performance is poor.

For our third region of interest, the right ventricular blood pool (Figure 8), all methods perform much better. Only sequentially training the model and *LwF* shows a visible performance loss compared to the upper bound of joint training. This is likely due to there only being two tasks. The domain differences caused by using different scanners may not be as significant as those introduced in the hippocampus datasets, where the patient populations differ.

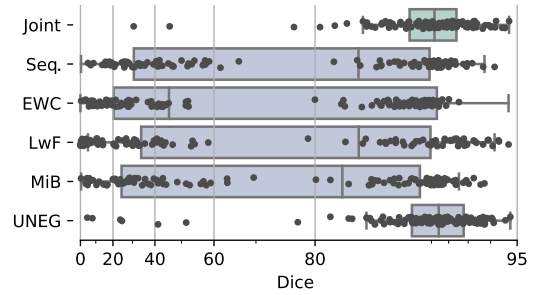


Figure 7: Dice scores of the final model state on test data from the three hippocampus segmentation datasets.

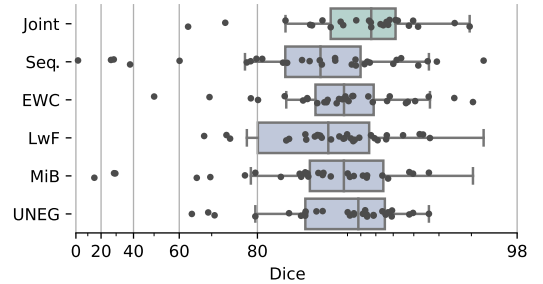


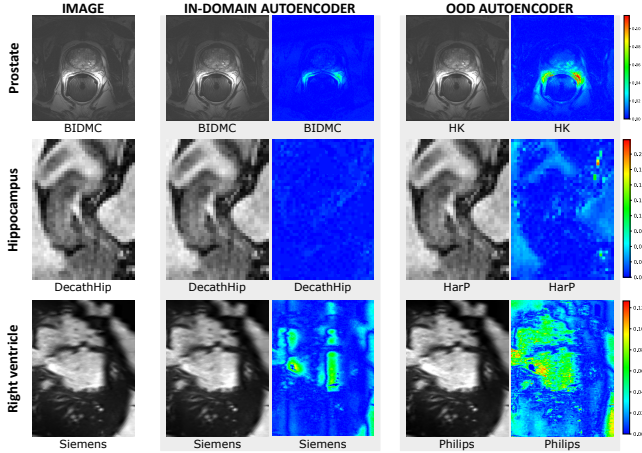
Figure 8: Dice for the task of right ventricle segmentation on test data from both datasets, Philips and Siemens.

6.2 Ablation Study

We compare several autoencoder options in Table 1. We calculate the mean Dice over all tasks, *BWT* as defined by González et al. [17], and how accurately the oracle properly identifies the domain. The first row shows the upper bound of using the ground truth task identities, as is possible in the *Domain Knowledge* scenario. The second row is the CNN autoencoder proposed by Aljundi et al. [2], which reconstructs *AlexNet* features. We then look into using the nnU-Net autoencoder (*UNEG*) and several other settings described in Section 5.4. While *AlexNet* z-CNN correctly identifies the domain for most prostate cases, it fails to do so for the hippocampus and cardiac examinations. *UNEG* instead achieves high accuracy for the three anatomies, and a high Dice across domains.

Table 1: Ablation study on the selection of the best autoencoder architecture for an oracle that infers task identity. We report the mean Dice, BWT [17] and the accuracy at selecting the “correct” task identity.

Method	Prostate			Hippocampus			Right ventricle		
	Dice $\uparrow$	BWT (%) $\uparrow$	Acc. $\uparrow$	Dice $\uparrow$	BWT (%) $\uparrow$	Acc. $\uparrow$	Dice $\uparrow$	BWT (%) $\uparrow$	Acc. $\uparrow$
Task identity	89.6 $\pm$ 2.0	-	-	90.6 $\pm$ 2.2	-	-	89.6 $\pm$ 1.5	-	-
AlexNet z-CNN	87.2 $\pm$ 6.6	-3.5 $\pm$ 6.0	84.2	73.3 $\pm$ 19.2	-23.9 $\pm$ 23.9	24.1	81.1 $\pm$ 9.9	-19.2 $\pm$ 0.0	50.0
UNEG	87.0 $\pm$ 6.5	-3.8 $\pm$ 5.9	68.4	89.0 $\pm$ 3.1	-2.6 $\pm$ 2.6	97.4	88.7 $\pm$ 2.3	-2.0 $\pm$ 0.0	90.0
MONAI	66.2 $\pm$ 29.7	-30.0 $\pm$ 36.2	31.6	65.0 $\pm$ 22.8	-29.4 $\pm$ 29.4	13.8	89.6 $\pm$ 1.5	-0.0 $\pm$ 0.0	100.0
CNN	80.3 $\pm$ 11.3	-11.5 $\pm$ 13.0	52.6	72.0 $\pm$ 28.3	-0.0 $\pm$ 0.0	56.0	87.5 $\pm$ 4.3	-5.6 $\pm$ 0.0	60.0
nnU-Net z-CNN	84.2 $\pm$ 7.0	-3.5 $\pm$ 6.1	52.6	60.7 $\pm$ 26.4	-15.0 $\pm$ 14.1	46.6	81.1 $\pm$ 9.9	-19.2 $\pm$ 0.0	50.0

**Figure 9: Qualitative evaluation of the reconstructions extracted by the “correct” vs. OOD autoencoder.**

6.3 Qualitative Evaluation of Reconstructions

We observe exemplary image reconstructions for the three anatomies in Figure 9, which shows the original image, the reconstruction produced by the autoencoder trained with data from the same domain, and the residual image highlighting the differences between the two. This is followed by the reconstruction made by an autoencoder from a different domain. When looking at the residuals, we notice that the in-distribution reconstructions are much better.

6.4 Discussion

The discussion on CL often revolves around *catastrophic forgetting*, the sharp drop in performance for domains seen in early training stages. Often, a trade-off is pursued between *rigidity* and *plasticity*, whereby the model preserves previous knowledge but can still adapt to changes in the environment. Yet there is a trivial way to circumvent the loss of performance in both directions: keeping a separate model per domain – stored after completing the respective training stage – and using the appropriate model at test time.

In this work, we introduce an evaluation workflow alongside a multi-model benchmark, a fair baseline that maintains one model per domain. The benchmark has no possibility of gaining useful information from data seen in later stages for an earlier domain, i.e. no *positive backward transfer* can occur. Yet related works show us that this is rarely the case in practice, which puts into question

whether the additional complexity of continual learning approaches that modify the training procedure is worth it.

Central to our evaluation scheme is the question of whether domain identity information is present during inference. Many CL methods assume so, which makes it possible to maintain a *multi-head* architecture where only some parameters are shared. Yet this assumption does not hold in many real-world settings, which compromises the applicability of the proposed methodologies.

For cases where identity labels are not known, we propose *UNEG* (*U-Net Expert Gate*), which trains one autoencoder per domain. The autoencoder replicates the architecture of the segmentation model – in our case, a patch-based nnU-Net. This makes use of the fact that the architecture and pre-preprocessing steps are already tuned to the particular input data and task. Our empirical evaluation exploring various autoencoder settings confirms that this is the most effective way to select the correct model at test time.

One limitation of our approach is that one additional model, namely the autoencoder, needs to be trained per stage. This can occur in parallel to training the segmenter but still implies the use of additional computation resources. In future work, we will explore more efficient *oracle* strategies.

7 Conclusion and Outlook

Many methods exist to prevent catastrophic forgetting for image classification, and several have been adapted with relative success to semantic segmentation. Yet **few methods achieve positive backward transfer**, i.e. while the model does not forget how to deal with data seen in early training stages, it also does not leverage information seen later on. In such cases, a multi-model solution would be preferable in clinical practice, where reliability is paramount and there is rarely a lack of persistent storage. In this work, we present a multi-model strategy alongside a fair evaluation framework for continual learning methods.

The proposed evaluation considers the fact that continual learning approaches often rely on receiving domain identity information during inference. This may not be the case in real-world dynamic environments, where metadata may be concealed for privacy reasons or the model must handle data from previously-unseen sources.

Continual learning methodologies are gaining a lot of attention from the diagnostic radiology community. Just like we are seeing more works that evaluate models on out-of-distribution data, we hope that training and evaluating models in a continual fashion and quantifying their backward transferability becomes common practice in the community.

References

- [1] Rahaf Aljundi, Francesca Babiloni, Mohamed Elhoseiny, Marcus Rohrbach, and Tinne Tuytelaars. 2018. Memory aware synapses: Learning what (not) to forget. In *Proceedings of the European Conference on Computer Vision (ECCV)*. 139–154.
- [2] Rahaf Aljundi, Punarjay Chakravarty, and Tinne Tuytelaars. 2017. Expert gate: Lifelong learning with a network of experts. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 3366–3375.
- [3] Rahaf Aljundi, Min Lin, Baptiste Goujaud, and Yoshua Bengio. 2019. Gradient based sample selection for online continual learning. *Advances in Neural Information Processing Systems* 32 (2019).
- [4] Bernard Ans and Stephane Rousset. 2000. Neural networks with a self-refreshing memory: knowledge transfer in sequential learning tasks without catastrophic forgetting. *Connection science* 12, 1 (2000), 1–19.
- [5] Michela Antonelli, Annika Reinke, Spyridon Bakas, Keyvan Farahani, Annette Kopp-Schneider, Bennett A Landman, Geert Litjens, Bjoern Menze, Olaf Ronneberger, Ronald M Summers, et al. 2022. The medical segmentation decathlon. *Nature communications* 13, 1 (2022), 4128.
- [6] Chaitanya Baweja, Ben Glocker, and Konstantinos Kamnitsas. 2018. Towards continual learning in medical imaging. *arXiv preprint arXiv:1811.02496* (2018).
- [7] Theodora S Brisimi, Ruidi Chen, Theofanie Mela, Alex Olshevsky, Ioannis Ch Paschalidis, and Wei Shi. 2018. Federated learning of predictive models from federated electronic health records. *International journal of medical informatics* 112 (2018), 59–67.
- [8] Victor M. Campello, Polyxeni Gkontra, Cristian Izquierdo, Carlos Martín-Isla, Alireza Sojoudi, Peter M. Full, Klaus Maier-Hein, Yao Zhang, Zhiqiang He, Jun Ma, Mario Parreño, Alberto Albiol, Fanwei Kong, Shawn C. Shadden, Jorge Corral Acero, Vaanathi Sundaresan, Mina Saber, Mustafa Elattar, Hongwei Li, Björn Menze, Firas Khader, Christoph Hauburger, Cian M. Scannell, Mitko Veta, Adam Carscadden, Kumaradevan Punithakumar, Xiao Liu, Sotirios A. Tsaftaris, Xiaoqiong Huang, Xin Yang, Lei Li, Xiaohai Zhuang, David Viladés, Martín L. Descalzo, Andrea Guala, Lucia La Mura, Matthias G. Friedrich, Ria Garg, Julie Lebel, Filipe Henriques, Mahir Karakas, Ersin Çavuş, Steffen E. Petersen, Sergio Escalera, Santi Seguí, José F. Rodríguez-Palomares, and Karim Lekadir. 2021. Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation: The M&Ms Challenge. *IEEE Transactions on Medical Imaging* 40, 12 (2021), 3543–3554. doi:10.1109/TMI.2021.3090082
- [9] M. Jorge Cardoso, Wenqi Li, Richard Brown, Nic Ma, Eric Kerfoot, Yiheng Wang, Benjamin Murrey, Andriy Myronenko, Can Zhao, Dong Yang, Vishwesh Nath, Yufan He, Ziyue Xu, Ali Hatamizadeh, Andriy Myronenko, Wentao Zhu, Yun Liu, Mingxin Zheng, Yucheng Tang, Isaac Yang, Michael Zephyr, Behrooz Hashemian, Sachidanand Alle, Mohammad Zalbagi Darestani, Charlie Budd, Marc Modat, Tom Vercauteren, Guotai Wang, Yiwen Li, Yipeng Hu, Yunguan Fu, Benjamin Gorman, Hans Johnson, Brad Genereaux, Barbaros S. Erdal, Vikash Gupta, Andres Diaz-Pinto, Andre Dourson, Lena Maier-Hein, Paul F. Jaeger, Michael Baumgartner, Jayashree Kalpathy-Cramer, Mona Flores, Justin Kirby, Lee A. D. Cooper, Holger R. Roth, Daguang Xu, David Bericat, Ralf Floca, S. Kevin Zhou, Haris Shuaib, Keyvan Farahani, Klaus H. Maier-Hein, Stephen Aylward, Prerna Dogra, Sebastien Ourselin, and Andrew Feng. 2022. MONAI: An open-source framework for deep learning in healthcare. doi:10.48550/ARXIV.2211.02701
- [10] Fabio Cermelli, Massimiliano Mancini, Samuel Rota Buló, Elisa Ricci, and Barbara Caputo. 2020. Modeling the background for incremental learning in semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 9233–9242.
- [11] Timothy J Draelos, Nadine F Miner, Christopher C Lamb, Jonathan A Cox, Craig M Vineyard, Kristofer D Carlson, William M Severa, Conrad D James, and James B Aimone. 2017. Neurogenesis deep learning: Extending deep networks to accommodate new classes. In *2017 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 526–533.
- [12] Sayna Ebrahimi, Mohamed Elhoseiny, Trevor Darrell, and Marcus Rohrbach. 2019. Uncertainty-guided Continual Learning with Bayesian Neural Networks. *arXiv preprint arXiv:1906.02425* (2019).
- [13] Sayna Ebrahimi, Franziska Meier, Roberto Calandra, Trevor Darrell, and Marcus Rohrbach. 2020. Adversarial continual learning. In *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XI*. Springer, 386–402.
- [14] Abdelrahman Elshahawy, Aneta Lisowska, Matthias Keicher, Joseph Henry, Paul Thomson, and Nassir Navab. 2020. Continual Class Incremental Learning for CT Thoracic Segmentation. In *Domain Adaptation and Representation Transfer, and Distributed and Collaborative Learning*. Springer, 106–116.
- [15] Ben Glocker, Robert Robinson, Daniel C Castro, Qi Dou, and Ender Konukoglu. 2019. Machine learning with multi-site imaging data: An empirical study on the impact of scanner effects. *arXiv preprint arXiv:1910.04597* (2019).
- [16] Siavash Golkar, Michael Kagan, and Kyunghyun Cho. 2019. Continual learning via neural pruning. *arXiv preprint arXiv:1903.04476* (2019).
- [17] Camila Gonzalez, Amin Ranem, Daniel Pinto dos Santos, Ahmed Othman, and Anirban Mukhopadhyay. 2022. Lifelong nnUNet: a framework for standardized medical continual learning. - (2022).
- [18] Camila González, Amin Ranem, Ahmed Othman, and Anirban Mukhopadhyay. 2022. Task-agnostic continual hippocampus segmentation for smooth population shifts. In *MICCAI Workshop on Domain Adaptation and Representation Transfer*. Springer, 108–118.
- [19] Ben Goodrich and Itamar Arel. 2014. Unsupervised neuron selection for mitigating catastrophic forgetting in neural networks. In *2014 IEEE 57th International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE, 997–1000.
- [20] Johannes Hofmanninger, Matthias Perkonig, James A Brink, Oleg Panykh, Christian Herold, and Georg Langs. 2020. Dynamic memory to alleviate catastrophic forgetting in continuous learning settings. In *Medical Image Computing and Computer Assisted Intervention—MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part II*. Springer, 359–368.
- [21] Fabian Isensee, Paul F Jaeger, Simon AA Kohl, Jens Petersen, and Klaus H Maier-Hein. 2021. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature methods* 18, 2 (2021), 203–211.
- [22] Neerav Karani, Krishna Chaitanya, Christian Baumgartner, and Ender Konukoglu. 2018. A lifelong learning approach to brain mr segmentation across scanners and protocols. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 476–484.
- [23] Ronald Kemker and Christopher Kanan. 2017. Fearnert: Brain-inspired model for incremental learning. *arXiv preprint arXiv:1711.10563* (2017).
- [24] Hyo-Eun Kim, Seungwook Kim, and Jaehwan Lee. 2018. Keep and learn: Continual learning by constraining the latent space for knowledge preservation in neural networks. In *Medical Image Computing and Computer Assisted Intervention—MICCAI 2018: 21st International Conference, Granada, Spain, September 16–20, 2018, Proceedings, Part I*. Springer, 520–528.
- [25] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. 2017. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences* 114, 13 (2017), 3521–3526.
- [26] Andreas Knoblauch, Edgar Körner, Ursula Körner, and Friedrich T Sommer. 2014. Structural synaptic plasticity has high memory capacity and can explain graded amnesia, catastrophic forgetting, and the spacing effect. *PLoS one* 9, 5 (2014), e96485.
- [27] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. 2012. ImageNet Classification with Deep Convolutional Neural Networks. In *Advances in Neural Information Processing Systems*, F. Pereira, C.J. Burges, L. Bottou, and K.Q. Weinberger (Eds.), Vol. 25. Curran Associates, Inc. <https://proceedings.neurips.cc/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf>
- [28] Jessie Kulaga-Yoskovitz, Boris C Bernhardt, Seok-Jun Hong, Tommaso Mansi, Kevin E Liang, Andre JW Van Der Kouwe, Jonathan Smallwood, Andrea Bernasconi, and Neda Bernasconi. 2015. Multi-contrast submillimetric 3 Tesla hippocampal subfield segmentation protocol and dataset. *Scientific data* 2, 1 (2015), 1–9.
- [29] Kibok Lee, Kimin Lee, Jinwoo Shin, and Honglak Lee. 2019. Overcoming catastrophic forgetting with unlabeled data in the wild. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 312–321.
- [30] Matthias Lenga, Heinrich Schulz, and Axel Saalbach. 2020. Continual Learning for Domain Adaptation in Chest X-ray Classification. In *Medical Imaging with Deep Learning*.
- [31] Tian Li, Anit Kumar Sahu, Ameet Talwalkar, and Virginia Smith. 2020. Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine* 37, 3 (2020), 50–60.
- [32] Zhizhong Li and Derek Hoiem. 2017. Learning without forgetting. *IEEE transactions on pattern analysis and machine intelligence* 40, 12 (2017), 2935–2947.
- [33] Quande Liu, Qi Dou, Lequan Yu, and Pheng Ann Heng. 2020. Ms-net: Multi-site network for improving prostate segmentation with heterogeneous mri data. *IEEE Transactions on Medical Imaging* (2020).
- [34] Arun Mallaya and Svetlana Lazebnik. 2018. Packnet: Adding multiple tasks to a single network by iterative pruning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 7765–7773.
- [35] Asato Matsumoto and Keiji Yanai. 2020. Continual learning of image translation networks using task-dependent weight selection masks. In *Pattern Recognition: 5th Asian Conference, ACPR 2019, Auckland, New Zealand, November 26–29, 2019, Revised Selected Papers, Part II*. Springer, 129–142.
- [36] Michael McCloskey and Neal J Cohen. 1989. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of learning and motivation*. Vol. 24. Elsevier, 109–165.
- [37] Patrick McClure, Charles Y Zheng, Jakub Kaczmarzyk, John Rogers-Lee, Satra Ghosh, Dylan Nielson, Peter A Bandettini, and Francisco Pereira. 2018. Distributed weight consolidation: A brain segmentation case study. In *Advances in Neural Information Processing Systems*. 4093–4103.
- [38] Umberto Michieli and Pietro Zanutigh. 2019. Incremental Learning Techniques for Semantic Segmentation. In *Proceedings of the IEEE International Conference on Computer Vision Workshops*. 0–0.
- [39] Ishan Misra, Abhinav Shrivastava, Abhinav Gupta, and Martial Hebert. 2016. Cross-stitch networks for multi-task learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 3994–4003.

- [40] Cuong V Nguyen, Yingzhen Li, Thang D Bui, and Richard E Turner. 2017. Variational continual learning. *arXiv preprint arXiv:1710.10628* (2017).
- [41] Giang Nguyen, Shuan Chen, Thao Do, Tae Joon Jun, Ho-Jin Choi, and Daeyoung Kim. 2020. Dissecting catastrophic forgetting in continual learning by deep visualization. *arXiv preprint arXiv:2001.01578* (2020).
- [42] Firat Ozdemir, Philipp Fuernstahl, and Orcun Goksel. 2018. Learn the new, keep the old: Extending pretrained models with new anatomy and images. In *Medical Image Computing and Computer Assisted Intervention—MICCAI 2018: 21st International Conference, Granada, Spain, September 16–20, 2018, Proceedings, Part IV 11*. Springer, 361–369.
- [43] Sinan Özgün, Anne-Marie Rickmann, Abhijit Guha Roy, and Christian Wachinger. 2020. Importance driven continual learning for segmentation across domains. In *Machine Learning in Medical Imaging: 11th International Workshop, MLMI 2020, Held in Conjunction with MICCAI 2020, Lima, Peru, October 4, 2020, Proceedings 11*. Springer, 423–433.
- [44] Oleg S Pinykh, Georg Langs, Marc Dewey, Dieter R Enzmann, Christian J Herold, Stefan O Schoenberg, and James A Brink. 2020. Continuous learning AI in radiology: implementation principles and early applications. *Radiology* (2020), 200038.
- [45] Amin Ranem, Mohamed Afham Mohamed Affal, Moritz Fuchs, and Anirban Mukhopadhyay. [n. d.]. UnCLE SAM: Unleashing SAM's Potential for Continual Prostate MRI Segmentation. In *Medical Imaging with Deep Learning*.
- [46] Amin Ranem, Camila González, Daniel Pinto dos Santos, Andreas M Bucher, Ahmed E Othman, and Anirban Mukhopadhyay. 2024. Continual atlas-based segmentation of prostate MRI. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 7563–7572.
- [47] Amin Ranem, Camila González, and Anirban Mukhopadhyay. 2022. Continual hippocampus segmentation with transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3711–3720.
- [48] Roger Ratcliff. 1990. Connectionist models of recognition memory: constraints imposed by learning and forgetting functions. *Psychological review* 97, 2 (1990), 285.
- [49] Hariharan Ravishankar, Rahul Venkataramani, Saihareesh Anamandra, Prasad Sudhakar, and Pavan Annangi. 2019. Feature transformers: privacy preserving lifelong learners for medical imaging. In *Medical Image Computing and Computer Assisted Intervention—MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13–17, 2019, Proceedings, Part IV 22*. Springer, 347–355.
- [50] Sylvestre-Alvise Rebuffi, Hakan Bilen, and Andrea Vedaldi. 2017. Learning multiple visual domains with residual adapters. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., Red Hook, NY, USA, 506–516.
- [51] Sylvestre-Alvise Rebuffi, Alexander Kolesnikov, Georg Sperl, and Christoph H Lampert. 2017. icarl: Incremental classifier and representation learning. In *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*. 2001–2010.
- [52] Andrei A Rusu, Neil C Rabinowitz, Guillaume Desjardins, Hubert Soyer, James Kirkpatrick, Koray Kavukcuoglu, Razvan Pascanu, and Raia Hadsell. 2016. Progressive neural networks. *arXiv preprint arXiv:1606.04671* (2016).
- [53] Hanul Shin, Jung Kwon Lee, Jaehong Kim, and Jiwon Kim. 2017. Continual learning with deep generative replay. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., Red Hook, NY, USA, 2990–2999.
- [54] Konstantin Shmelkov, Cordelia Schmid, and Karteek Alahari. 2017. Incremental learning of object detectors without catastrophic forgetting. In *Proceedings of the IEEE International Conference on Computer Vision*. 3400–3409.
- [55] Santiago Silva, Boris A Gutman, Eduardo Romero, Paul M Thompson, Andre Altmann, and Marco Lorenzi. 2019. Federated learning in distributed medical databases: Meta-analysis of large-scale subcortical brain data. In *2019 IEEE 16th international symposium on biomedical imaging (ISBI 2019)*. IEEE, 270–274.
- [56] Gido M van de Ven, Tinne Tuytelaars, and Andreas S Tolias. 2022. Three types of incremental learning. *Nature Machine Intelligence* 4, 12 (2022), 1185–1197.
- [57] Karin van Garderen, Sebastian van der Voort, Fatih Incekara, Marion Smits, and Stefan Klein. 2019. Towards continuous learning for glioma segmentation with elastic weight consolidation. *MIDL* (2019).
- [58] Rahul Venkataramani, Hariharan Ravishankar, and Saihareesh Anamandra. 2019. Towards continuous domain adaptation for medical imaging. In *2019 IEEE 16th International Symposium on Biomedical Imaging (ISBI 2019)*. IEEE, 443–446.
- [59] Yu-Xiong Wang, Deva Ramanan, and Martial Hebert. 2017. Growing a brain: Fine-tuning by increasing model capacity. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2471–2480.
- [60] Laura EM Wisse, Ana M Daugherty, Rosanna K Olsen, David Berron, Valerie A Carr, Craig EL Stark, Robert SC Amaral, Katrin Amunts, Jean C Augustinack, Andrew R Bender, et al. 2017. A harmonized segmentation protocol for hippocampal and parahippocampal subregions: Why do we need one and what are the key goals? *Hippocampus* 27, 1 (2017), 3–11.
- [61] Zuxuan Wu, Xin Wang, Joseph E Gonzalez, Tom Goldstein, and Larry S Davis. 2019. Ace: Adapting to changing environments for semantic segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. IEEE, Seoul, Korea (South), 2121–2130.
- [62] Ning Yu, Larry S. Davis, and Mario Fritz. 2019. Attributing Fake Images to GANs: Learning and Analyzing GAN Fingerprints. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.